The Rajasthani Schools of Painting 2



The term 'Rajasthani Schools of Painting' pertains to the schools of painting that prevailed in the princely kingdoms and *thikanas* of what roughly constitutes Rajasthan and parts of Madhya Pradesh in the present time, such as Mewar, Bundi, Kota, Jaipur, Bikaner, Kishangarh, Jodhpur (Marwar), Malwa, Sirohi and other such principalities largely between the sixteenth and early nineteenth centuries.

Scholar Anand Coomaraswamy in 1916 coined the term 'Rajput Paintings' to refer to these as most rulers and patrons of these kingdoms were Rajputs. He, specifically, coined it to categorise and differentiate this group from the much known Mughal School of Painting. Therefore, Malwa, comprising princedoms of Central India, and the Pahari Schools that comprises the *pahari* or mountainous Himalayan region of north-western India was also in the ambit of Rajput Schools. For Coomaraswamy, the nomenclature represented the indigenous tradition of painting prevalent in the mainland before the conquest by the Mughals. Studies in Indian paintings have come a long way since then and the term 'Rajput Schools' is obsolete. Instead, specific categories, such as Rajasthani and Pahari are employed.

Though separated by short distances, the pictorial styles that emerged and evolved in these kingdoms were significantly diverse in terms of either execution—fine or bold; preference of colours (brilliant or gentle); compositional elements (depiction of architecture, figures and nature); modes of narration; affinity for naturalism—or had emphasis on extreme mannerism.

Paintings were painted on waslis—layered, thin sheets of handmade papers glued together to get the desired thickness. The outline was sketched on waslis in black or brown followed by colours fixed therein by brief notations or sample patches. Colour pigments were predominantly obtained from minerals and precious metals like gold and silver that were mixed with glue as the binding medium. Camel and squirrel

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hair were used in brushes. On completion, the painting was burnished with an agate to lend it a uniform sheen and an appealing resplendence.

The painting activity was a kind of teamwork, with the master artist composing and doing preliminary drawings, followed by pupils or experts of colouring, portraiture, architecture, landscape, animals, etc., taking over and doing their bit, and finally, the master artist putting the finishing touches. The scribe would write the verse in the space left for the one.

Themes of Paintings – An Overview

By the sixteenth century, Vaishvanism in the cults of Rama and Krishna had become popular in many parts of western, northern and central India as part of the Bhakti movement that had swept the entire Indian subcontinent. Krishna had a special appeal. He was not only worshipped as God but also as an ideal lover. The notion of 'love' was cherished as a religious theme, where a delightful synthesis of sensuousness and mysticism was perceived. Krishna was perceived as the

Krishna and gopis in the forest, Gita Govinda, Mewar, 1550, Chhatrapati Shivaji Maharaj Vastu Sangrahalaya, Mumbai



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creator from whom all creation was a sportive emanation, and Radha, the human soul who led to offer herself to God. The soul's devotion to the deity is pictured by Radha's self-abandonment to her beloved Krishna epitomised in *Gita Govinda* paintings.

Composed in the twelfth century by Jayadeva, who is believed to have been the court poet of Lakshmana Sen of Bengal, *Gita Govinda*, the 'Song of the Cowherd', is a lyrical poem in Sanskrit, evoking *shringara rasa*, portraying the mystical love between Radha and Krishna through worldly imageries. Bhanu Datta, a Maithil Brahmin who lived in Bihar in the fourteenth century, composed another favourite text of artists, *Rasamanjari*, interpreted as the 'Bouquet of Delight'. Written in Sanskrit, the text is a treatise on *rasa* and deals with the classification of heroes (*nayakas*) and heroines (*nayikas*) in accordance with their age—*baal*, *taruna* and *praudha*; physiognomic traits of appearance, such as *padmini*, *chitrini*, *shankhini*, *hastini*, etc., and emotional states, such as *khandita*, *vasaksajja*, *abhisarika*, *utka*, etc. Though Krishna is not mentioned in the text, painters have introduced him as the archetypal lover.

Rasikapriya, translated as 'The Connoisseur's Delight', is replete with complex poetic interpretations and was composed to incite aesthetic pleasure to elite courtiers. Composed in Brajbhasha by Keshav Das, the court poet of Raja Madhukar Shah of Orchha in 1591, *Rasikapriya* explores various emotive states, such as love, togetherness, jilt, jealousy, quarrel and its aftermath, separation, anger, etc., that are common between lovers represented through the characters of Radha and Krishna.

Kavipriya, another poetic work by Keshav Das, was written in the honour of Rai Parbin, a celebrated courtesan of Orchha. It is a tale of love and its tenth chapter evocatively titled *Baramasa* engages with the most enduring climactic description of the 12 months of the year. While illustrating the daily life of people in different seasons and alluding to festivals falling therein, Keshav Das describes how the *nayika* prevails upon the *nayaka* not to leave her and proceed on a journey.

Bihari Satsai, authored by Bihari Lal, constituting 700 verses (*satsai*), is composed in the form of aphorisms and moralising witticism. It is largely held that he composed the *Satsai* around 1662 while he was at the court of Jaipur

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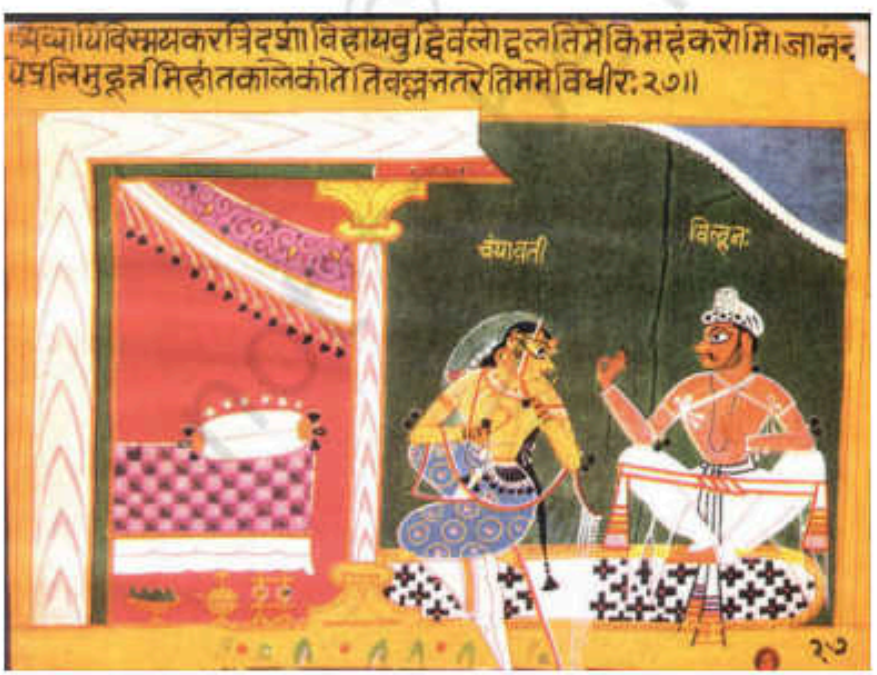
working for Mirza Raja Jai singh as the patron's name appears in several verses of the *Satsai*. The *Satsai* has been largely painted at Mewar and less frequently in the Pahari School.

Ragamala paintings are pictorial interpretations of *ragas* and *raginis*.

Ragas are traditionally envisioned in divine or human form in romantic or devotional contexts by musicians and poets. Each *raga* is associated with a specific mood, time of the day and season. *Ragamala* paintings are arranged in albums invariably containing 36 or 42 folios, organised in the format of families. Each family is headed by a male *raga*, having six female consorts called *raginis*. The six main *ragas* are *Bhairava*, *Malkos*, *Hindol*, *Dipak*, *Megha* and *Shri*.

Bardic legends and other romantic tales, such as *Dhola-Maru*, *Sohni-Mahiwal*, *Mrigavat*, *Chaurpanchashika* and *Laurchanda* just to mention a few were other favourite themes. Texts, such as the *Ramayana*, *Bhagvata Purana*, *Mahabharata*, *Devi Mahatmya* and the like were favourites with all schools of painting.

Moreover, a large number of paintings record *darbar* scenes and historic moments; depict hunting expeditions, wars and victories; picnics, garden parties, dance and music performances; rituals, festivals and wedding processions; portraits of kings, courtiers and their families; city views; birds and animals.



Chaurpanchasika, Mewar, 1500, N. C. Mehta Collection, Ahmedabad, Gujarat

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The Mughal School of Miniature Painting 3

Mughal painting is the style of miniature painting that developed in the northern Indian subcontinent in the sixteenth century and continued till the mid–nineteenth century. It is known for its sophisticated techniques and diverse range of subjects and themes. The Mughal miniature painting inspired and resonated in subsequent schools and styles of Indian painting, thereby, confirming a definite position for the Mughal style within the Indian school of paintings.

The Mughals were patrons of various art forms. Every Mughal successor, based on his taste and preferences, contributed towards enhancing the status of art, viz., calligraphy, painting, architecture, bookmaking, book illustration projects, etc. They took keen interest in artists' ateliers and nurtured unprecedented new styles that heightened and accelerated the existing art scenario of India. Therefore, for understanding the Mughal Painting, the political history and genealogy of the Mughal dynasty is often taken into account.

Influences on Mughal Painting

The Mughal style of miniature painting was responsible for the amalgamation of indigenous themes and styles along with Persian and later European themes and styles. The arts of this period reflect a synthesis of foreign influences and indigenous flavour. The peak of Mughal painting presented a highly sophisticated blend of the Islamic, Hindu and European visual culture and aesthetics. Given this diverse yet inclusive nature, the affluence of the artworks produced in India during this period surpasses the conventional and indigenous Indian and Iranian painting of that time. The significance of this style lies in the purpose and efforts of its patrons and the unmatched skill of its artists. Together, they envisaged and expressed a congregation of tastes, philosophies and faiths by their extraordinary visual language.

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In the Mughal courts, arts became more formalised as there were workshops and many artists were brought from Iran, which resulted in a harmonious blend of Indo-Iranian styles, especially during its early years. This celebrated eminence in Mughal art was possible only due to its distinctive character of assimilating and engaging artists of both Indian and Iranian origins, who contributed towards making, and further elevating the artistic paradigm of the Mughal style.

The Mughal atelier consisted of calligraphers, painters, gilders and binders. Paintings recorded and documented significant events, personalities and interests of the emperors. These were meant to be seen by the royals only. The paintings were made to suit the sensitivity of the royals or often made as intellectual stimulation. The paintings were a part of manuscripts and albums.

The tradition of art and painting had rich historical roots in India about which we have already learned in the previous chapters. The celebrated Mughal idiom that developed on the Indian soil should be understood as a consequence of interaction of various schools, including the pre–Mughal and contemporaneous art schools of India and Persia. Thus, the Mughal style did not grow in vacuum. It was nurtured by direct interaction with other art forms and schools that already existed. The indigenous Indian and Mughal painting styles coexisted, assimilated the influences and various native talents in different ways.

The pre–Mughal and parallel indigenous schools of paintings in India had their strong distinct style, aesthetics and purpose. The indigenous Indian style emphasised on flat perspective, strong use of lines, vivid colour palette, and bold modelling of figures and architecture. The Mughal style offered subtlety and finesse, portrayed almost three-dimensional figures and created optical reality. The royal court scenes, portraits, depiction of accurate flora and fauna were some of the favourite themes of the Mughal artists. Thus, the Mughal painting ushered in a new style and sophistication to the Indian arts of that time.

The Mughal patrons contributed to the proliferation of the Mughal style of painting with their distinct artistic preferences, choice of subjects, philosophies and aesthetic sensibilities. In the following segment of this chapter, we shall learn about the development of Mughal miniature painting in chronological order.

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Early Mughal Painting

In 1526, Babur, the first Mughal emperor, came from present day Uzbekistan and was a descendent of Emperor Timur and Chatghtai Turk. With this, he blended the cultural background and aesthetic sensibilities of Persia and Central Asia. Babur had a dynamic taste for a variety of arts. He is reputed to be a man of letters and a keen patron of art, manuscripts, architecture, gardening, etc. Babur's detailed accounts in *Baburnama*, his autobiography, are narratives of the emperor's political carrier and artistic passion. *Baburnama* reflects the love and fondness that Babur had as an outsider for the Indian land and ecology. With his fervour for detailed writing, Babur established a tradition of keeping memoirs, a practice which was followed by his successors in India. The books and albums that were produced in the royal ateliers were not only calligraphed but also painted. These books of value were preserved and passed on to members of the royal family or gifted to those perceived as deserving. Babur had a keen eye for portraiture and this is also recorded in his memoirs. Among the artists, who find mention in Babur's memoirs is Bihzad. Bihzad's work was



Princes of the House of Timur, Abd us Samad, 1545–50, British Museum, London

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dainty but he did not draw the faces well; he used to greatly lengthen the double chin (*ghab-ghab*); and drew bearded faces admirably. Bihzad was a master artist from the Persian school of painting, Herat (now in present day Afghanistan), and was known for his sophisticated compositions and colour tints. Also, Shah Muzaffar finds a mention as a painter, who Babur thought excelled in the representation of hairstyle. Although Babur spent little time on the Indian soil and passed away soon after his arrival, his successors made the country their own and became a part of the Indian lineage.

Babur was succeeded by his son Humayun in 1530, who unfortunately fell prey to political unrest, and his life took many unexpected turns. Dethroned by an Afghan, Sher Khan (Sher Shah), Humayun took refuge in the court of the Safavid Persian ruler, Shah Tahmasp. Although inglorious for his political career, this was fortunate for the startling turn that the art of manuscript and painting under him took as a result of his stay in Safavid. It was during his exile in Shah Tahmasp's court that Humanyun witnessed the magnificent artistic tradition of miniature paintings and manuscripts. He was thrilled to see the skillful artists in practice, creating splendid works of art for Shah Tahmasp. With the assistance of Shah Tahmasp, Humayun established his court in Kabul in 1545. Humayun increasingly identified himself with a political and cultural agenda for his dynastic empire that was eclectic and assimilative. Impressed by the artists and with an ambition to recreate such art workshops in India, Humayun brought back with him the master artists when he regained power in India. He invited two Persian artists—Mir Sayyid Ali and Abd us Samad to establish a studio in his court and carry out royal paintings. It is important to note here that both the artists were famous and respected particularly for their skills in the art of portraiture.



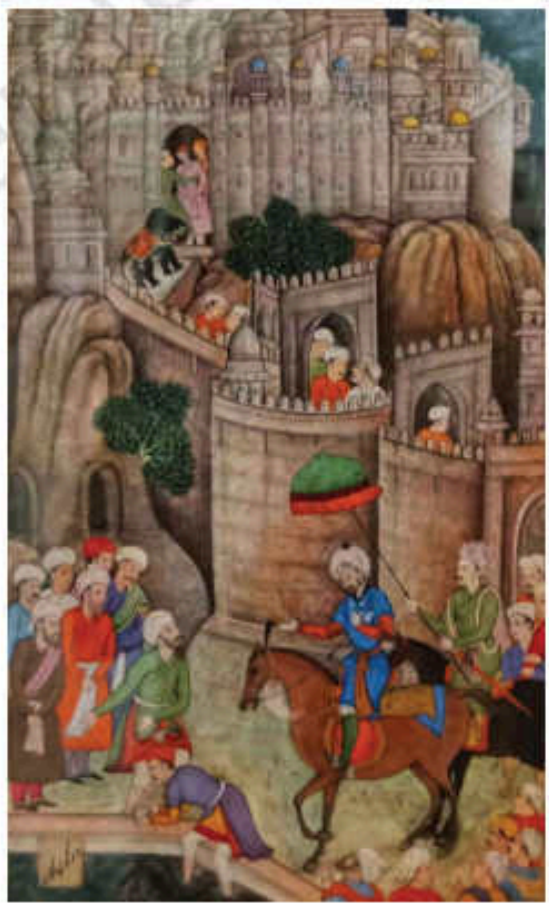
Tutinama: The Girl and the Parrot, 1580–1585, Chester Beatty Library, Dublin

A bibliophile of discerning sensitivity, Humayun's rule began a period of intense patronage for the art of painting and calligraphy. From his period, we get clear visual and textual documents that testify an active

interest in building an artistic repertoire and an imperial atelier. This is an indication of Humayun's artistic taste and helps us form a picture of Humayun as a connoisseur and an aesthete. He founded the *Nigaar Khana* (painting workshop), which was also a part of his library. Not much is known about the size and composition of Humayun's workshop in India. However, it is known that he started the project of illustration of *Hamza Nama* that was continued by his son and successor Akbar.

When we look at an extraordinary Mughal painting from the early period, *Princes of the House of Timur* (1545–50), probably by Safavid artist, Abd us Samad, in opaque watercolour on cotton, we are surprised by its size and complex structure and display of historical portraits. A prized possession of the imperial family, it has portraits that were painted over the original to register the portraits of successive members of the Mughal dynasty. So, visible in their physical likeness are portraits of Akbar, Jahangir and Shah Jahan painted subsequently over those painted during the reign of Humayun.

Babur inspecting the fort of Gwalior, Bhure, Baburnama, 1598, National Museum, New Delhi



Open air painting with trees and blossoms, and royal merrymaking, which depicts the ancestral members of the Mughal dynasty were followed after Humayun, who was the patron of this kind of artwork. The format, theme, figures and colour palette are remarkably Persian. There is indeed at this point we may state that there is no particular dominant element having Indian inspiration. But soon, this vocabulary changes to accommodate the growing and peculiar Mughal sensibility and distinct imperial taste.

The tradition and fascination for painting started by Humayun was carried forward by his illustrious son Akbar (1556–1605). Abul Fazal, the court historian of Akbar, writes about Akbar's passion for arts. He records that more than a hundred artists were employed in the royal atelier. This included the most skilled Persian and indigenous Indian artists of that time. This integrated composition of Indo-Persian artists led to the development of a unique style in this period. These artists together undertook ambitious projects that established new artistic

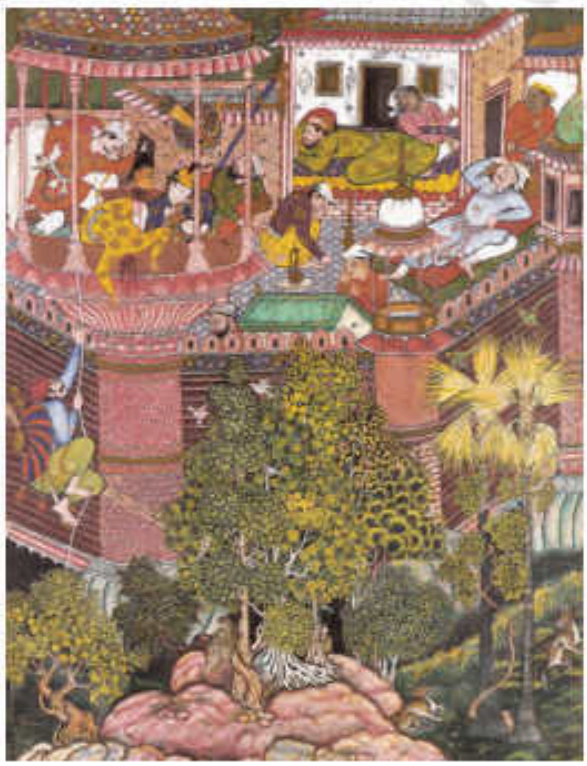
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standards with regard to visual language, as well as, subject matter. Akbar, believed to be suffering from dyslexia (a condition where a person finds difficulty in reading), laid great emphasis on illustration of manuscripts. It was under his patronage that several seminal projects of translation and illustration of manuscripts were carried out.

The earliest of his projects is the continuation of his father's artistic legacy of *Hamza Nama*, an illustrated account of the heroic deeds of Hamza, the uncle of Prophet Muhammad. Akbar took delight in hearing the stories of Hamza, a character much loved in the Middle East popular and intellectual circles, which were read out aloud by a professional narrator. Simultaneously, the corresponding folios and painted *Hamza Nama* narrative was held for a clear view. The emperor took great interest in both pictorial narrative, as well as, recitation of *Hamza Nama*. Because of the peculiar function of these paintings, their format is large. The base surface is cloth with paper at the back, on which the narrative text is written to help the narrator and the technique applied is gouache, which is water-based and in opaque colours.

One realises that Mughal paintings were a teamwork by a group of artists, who could be inspired by a number of artistic traditions. The immediate natural surroundings became the resource from which the images of flora and fauna were derived and painted. The painted folios of *Hamza Nama* are scattered all over the world and housed in various collections. It is recorded to have consisted of 14 volumes with 1400 illustrations and took nearly 15 years to be completed. The suggested date of this magnificent project is 1567–1582 and was completed under the supervision of two Persian masters—Mir Sayyid Ali and Abd us Samad.

Spies of Hamza attack the City of Kaymar, 1567–1582, Museum of Applied Arts, Vienna



In *Hamza Nama* painting, *Spies Attack the City of Kaymar* (1567–82), the space is sharply cut and divided so as to facilitate visual reading of the narrative. Too much of action is happening and the vibrant colours are of great use here to energise the unfolding of this story, wherein, the spies of Hamza attack the city of Kaymar. A strong outer line defines the foliage and other forms. The faces are largely

seen in profile. However, three–fourth faces are also shown. The rich intricate patterns on the floor, columns and canopy are from Persian sources as also the four-limbed animals and rocks. Trees and creepers indicate Indian source as also the rich palette of pure yellows, reds and browns.

Akbar envisioned cultural integration and commissioned translation of several revered Hindu texts. He commissioned translation and illustration of revered Sanskrit texts into Persian. The Persian translation and illustrated version of the Hindu epic *Mahabharata* done this period came to be known as *Razm Nama*. This was completed in 1589 under the supervision of master artist Daswant. This manuscript was scribed in ornate calligraphy and contained 169 paintings. The translation and illustration of the *Ramayana* were also carried out around this time. Artists like Govardhan and Miskin were celebrated for their visuals of court scenes. *Akbar Nama*, an extraordinary manuscript, containing a detailed account of Akbar's political and personal life, was one of the most expensive projects undertaken by Akbar.

Akbar personally engaged with the artists, and supervised and evaluated the artworks. The Mughal painting under Akbar's patronage depicted a variety of subjects, including, detailed political conquests, seminal court scenes, secular texts, portraits of important men along with Hindu mythologies, and Persian and Islamic themes. Akbar's fascination for Indian scriptures and regard for India made him one of the most popular emperors of the country.

In most of the paintings, produced from the time that the Europeans were in contact with the court of Akbar, we can see an increasing preference for a category of naturalism adapted to compliment the growing diversity in medieval India. *Madonna and Child* (1580) done in opaque watercolour on paper is an important early work of the Mughal School of Painting in this context. Madonna, here, is an extraordinary theme, which brings the Byzantine art, the European classical and its Renaissance to the Mughal atelier, where it is translated and transformed into an entirely different visual experience. Virgin Mary is draped in a classical manner. The attachment displayed between the mother and the child was inspired by the humanist interpretation in the European Renaissance art. The physiology of the child, certain details like the fan and jewellery completely integrated the work to an Indian milieu.

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Madonna and Child, Basawan, 1590, San Diego Museum of Art, California

Inspired by Akbar's interest in arts, many sub-imperial courts absorbed this passion and several great works of art were produced for aristocratic families, who tried to copy the Mughal court atelier taste and produced works that present distinctive subjects and visual preferences in regional flavour.

Akbar had formalised the Mughal miniature style and set standards, which were further taken to new heights by his son Jahangir (1605–1627). Prince Salim, (Jahangir) showed interest in art from an early age. Unlike his father Akbar, who commissioned paintings and manuscripts of politically and religiously significant aspects, Prince Salim had a curious taste and encouraged delicate observations and fine details.

Jahangir employed Aqa Riza, a well-known Iranian painter and his son Abul Hasan to achieve unparalleled sophistication in painting. Despite the formalised and established imperial atelier of Akbar, the keen patron in Jahangir rebelled to set his own atelier alongside his father's. Prince Salim came to be known as Jahangir—the World Seizer once he acquired the Mughal throne after his return from Allahabad. *Tuzuk-i-Jahangiri*, the memoirs of Jahangir, tells about his great interest in arts and his efforts of achieving scientific correctness in the rendering flora and fauna that interested the Emperor the most. Under his patronage, the Mughal painting achieved naturalism and scientific accuracy of the highest degree. The curiosity and wonderment that the Emperor had for nature and people around him is reflected in the works that he commissioned.

In contrast to the Akbar's atelier, where the works were mass produced, Jahangir's atelier gave preference to a lesser number and better quality of artworks produced by a single master artist. The *Muraqqas* individual paintings to be mounted

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in albums became popular under Jahangir's patronage. The margins of the paintings were highly illuminated in gold and embellished with flora, fauna, and often poised human figures. The war scenes, portraits, narrative and storytelling prevalent in Akbar's style were overtaken by minute details and refined rendering of lavish court scenes, aristocracy, royal personalities, as well as, character traits, and distinctiveness of flora and fauna.

Jahangir was presented with paintings and decorative objects, depicting high arts from Europe, as gifts from the Europeans, who visited his court. With such a contact with the English Crown, Jahangir's fascination for European art and theme prompted him to have more such works in his collection. Many celebrated religious Christian themes were also produced in the royal atelier of Jahangir. Given this cultural and artistic exposure, European art sensibilities started making their way into the prevalent Indo-Iranian style, thereby, making the Jahangir school of art more impressive and vibrant. The spatial depth of the composition and naturalistic representation of life reflect the high benchmarks that the sensitive patron created for art during his lifetime. The artists of the Mughal atelier creatively assimilated the three styles—indigenous, Persian and European, making the Mughal Art School a melting pot of vibrant styles of its time yet very distinct in its own way.



A Prince and a Hermit, folio from Diwan of Amir Shahi, 1595, Aga Khan Museum, Canada

Jahangir in Darbar from *Jahangirnama* (now, dispersed), attributed to Abul Hasan and Manohar (1620) is an excellent painting. Jahangir is at the highest level in the centre, where the eyes move immediately from his figured frame to stunning white pillars surrounded by sparkling clear colours and brilliantly framed overhead canopy. On the right side,

The Pahari Schools of Painting 5

Pahari denotes 'hilly or mountainous' in origin. Pahari Schools of Painting includes towns, such as Basohli, Guler, Kangra, Kullu, Chamba, Mankot, Nurpur, Mandi, Bilaspur, Jammu and others in the hills of western Himalayas, which emerged as centres of painting from seventeenth to nineteenth century. Beginning at Basohli with a coarsely flamboyant style, it blossomed into the most exquisite and sophisticated style of Indian painting known as the Kangra School, through the Guler or pre-Kangra phase.

Unlike the distinguishing stylistic features of Mughal, Deccani and Rajasthani Schools, Pahari paintings demonstrate challenges in their territorial classification.

Though all the above centres crafted precisely individualistic characteristics in painting (through the depiction of nature, architecture, figural types, facial features, costumes, preference for particular colours and such other things), they do not develop as independent schools with distinctive styles. Paucity of dated material, colophons and inscriptions also prevent informed categorisation.

The emergence of the Pahari School remains unclear, though scholars have cautiously proposed theories concerning its beginning and influences. It is widely accepted that Mughal and Rajasthani styles of paintings were known in the hills probably through examples of Provincial Mughal style and family relationships of hill Rajas with the royal courts of Rajasthan. However, the flamboyantly bold Basohli-like style is, generally, understood to be the earliest prevailing pictorial language. B. N. Goswamy, one of the most significant scholars of the Pahari Schools of Painting, has attributed the shaping of Pahari style from



Krishna steals butter, Bhagvata Purana, 1750, N. C. Mehta Collection, Ahmedabad, Gujarat, India

the simplicity of Basohli to poetic lyricism and refinement of Kangra to the ingenuity of a family of artists through his scholarly approach of family as the basis of style. His central argument is that the family of Pandit Seu (Shiv) was chiefly responsible for the course of Pahari paintings. He argues that identifying Pahari paintings on the basis of regions could be misleading as political boundaries were always fluid. This argument is also true for Rajasthani schools as attribution merely by regions creates vagueness and several disparities remain unexplained. Hence, if a family of artists is considered as the style bearer, justification of multiple strands of a style can be accommodated within the same region and school.

Scholars agree that in the early eighteenth century, the style of the Seu family and others conformed to the Basohli idiom. However, from middle of the eighteenth century, the style transformed through a pre-Kangra phase, maturing into the Kangra style. This abrupt transformation in style and beginning of experimentation, which gave rise to varied stylistic idioms related to different Pahari centres, is largely ascribed to responses by various artist families and paintings (especially, the Mughal style) that were introduced in the Pahari kingdoms. This sudden arrival of paintings, which might have been introduced through rulers, artists, traders or any such agency or event, impacted local artists and profoundly influenced their painting language.

Rama and Sita in the forest, Kangra, 1780, Douglas Barrett Collection, UK



Most scholars, now, dispute the earlier hypothesis that the sudden change was caused and initiated by the migration of artists from the Mughal atelier.

For Goswamy, it was the naturalism in these paintings that appealed to the sensibilities of Pahari artists.

Compositions, worked out from a relative point of view, show some paintings with decorated margins. Themes that included recording the daily routine or important occasions from the lives of kings, creation of new prototype for female form and an idealised face, are all associated with this newly emerging style that gradually matures to the Kangra phase.

Basohli School

The first and most dramatic example of work from the hill states is from Basohli. From 1678 to 1695, Kirpal Pal, an enlightened prince, ruled the state. Under him, Basohli developed a distinctive and magnificent style. It is characterised by a strong use of primary colours and warm yellows—filling the background and horizon, stylised treatment of vegetation and raised white paint for imitating the representation of pearls in ornaments. However, the most significant characteristic of Basohli painting is the use of small, shiny green particles of beetle wings to delineate jewellery and simulate the effect of emeralds. In their vibrant palette and elegance, they share the aesthetics of the Chaurpanchashika group of paintings of Western India.

The most popular theme of Basohli painters was the *Rasamanjari* of Bhanu Datta. In 1694–95, Devida, a *tarkhan* (carpenter–painter), did a magnificent series for his patron Kirpal Pal. *Bhagvata Purana* and *Ragamala* were other popular themes. Artists also painted portraits of local kings with their consorts, courtiers, astrologers, mendicants,

Rasamanjari, Basohli, 1720, British Museum, London, UK



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The Bengal School and Cultural Nationalism

6

Company Painting

Art in India had a different purpose prior to the coming of the British. It could be seen as statues on temple walls, miniature paintings that often illustrated manuscripts, decoration on the walls of mud houses in villages, among many other examples. With the colonial rule around the eighteenth century, the English were charmed by different manners and customs of people of all ranks, tropical flora and fauna, and varying locales. Partly for documentation and partly for artistic reasons, many English officers commissioned local artists to paint scenes around them to get a better idea of the natives. The paintings were largely made on paper by local artists, some of whom had migrated from erstwhile courts of Murshidabad, Lucknow or Delhi. To please their new patrons, they had to adapt their traditional way of painting to document the world around them. This meant that they had to rely more on close observation, a striking feature of the European art, rather than memory and rule books, as seen in traditional art. It is this mixture of traditional and European style of painting that came to be known as the Company School of Painting. This style was not only popular among the British in India but even in Britain, where albums, consisting a set of paintings were much in demand.



Ghulam Ali Khan, Group of Courtesans, Company Painting, 1800–1825. San Diego Museum of Art, California, USA

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Raja Ravi Varma,
Krishna as envoy,
1906. NGMA, New Delhi, India

Raja Ravi Varma

This style declined with the entry of photography in India in the mid–nineteenth century as camera offered a better way of documentation. What, However, flourished in the art schools set up by the British was the academic style of oil painting that used a European medium to depict Indian subject matter. The most successful examples of this type of painting were found away from these art schools. They are best seen in the works produced by self-taught artist, Raja Ravi Varma of the Travancore Court in Kerala. By imitating copies of European paintings popular in Indian palaces, he mastered the style of academic realism and used it to depict scenes from popular epics like the *Ramayana* and *Mahabharata*. They became so popular that many of his paintings were copied as oleographs and were sold in market. They even entered people's homes as calendar images. With the rise of nationalism in India by the end of the nineteenth century, this academic style embraced by Raja Ravi Varma came to be looked down upon as foreign and too western to show Indian myths and history. It is amidst such nationalist thinking that the Bengal School of Art emerged in the first decade of the twentieth century.

The Bengal School

The term 'Bengal School of Art' is not fully accurate. It is true that the first move to create a modern, nationalist school happened in Bengal but it was not restricted to this region alone. It was an art movement and a style of painting that originated in Calcutta, the centre of British power, but later influenced many artists in different parts of the country, including Shantiniketan, where India's first national art school was founded. It was associated with the nationalist movement (*Swadeshi*) and spearheaded by Abanindranath Tagore (1871–1951). Abanindranath enjoyed the support of British administrator and principal of the Calcutta School of Art, E. B. Havell (1861–1934). Both Abanindranath and Havell were critical of colonial Art Schools and the manner in which European taste in art was being imposed on Indians. They firmly believed in creating a new type of painting that

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was Indian not only in subject matter but also in style. For them, Mughal and Pahari miniatures, for example, were more important sources of inspiration, rather than either the Company School of Painting or academic style taught in the colonial Art Schools.

Abanindranath Tagore and E. B. Havell

The year 1896 was important in the Indian history of visual arts. E. B. Havell and Abanindranath Tagore saw a need to Indianise art education in the country. This began in the Government Art School, Calcutta, now, Government College of Art and Craft, Kolkata. Similar art schools were established in Lahore, Bombay and Madras but their primary focus was on crafts like metalwork, furniture and curios. However, the one in Calcutta was more inclined towards fine arts. Havell and Abanindranath Tagore designed a curriculum to include and encourage technique and themes in Indian art traditions. Abanindranath's *Journey's End* shows the influence of Mughal and Pahari miniatures, and his desire to create an Indian style in painting.

Art historian Partha Mitter writes, "The first generation of the students of Abanindranath engaged in recovering the lost language of Indian art." To create awareness that modern Indians could benefit from this rich past, Abanindranath was the main artist and creator of an important journal, *Indian Society of Oriental Art*. In this manner, he was also the first major supporter of *Swadeshi* values in Indian art, which best manifested in the creation of Bengal School of Art. This school set the stage for the development of modern Indian painting. The new direction opened by Abanindranath was followed by many younger artists like Kshitindranath Majumdar (*Rasa-Lila*) and M. R. Chughtai (*Radhika*).

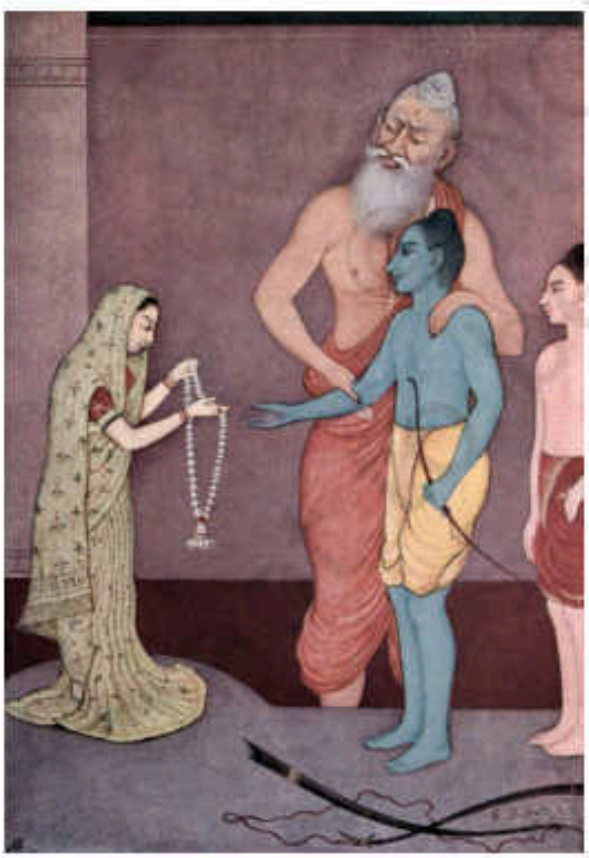
Shantiniketan — Early modernism

Nandalal Bose, a student of Abanindranath Tagore, was invited by poet and philosopher Rabindranath Tagore to head the painting department in the newly established Kala Bhavana. Kala Bhavana was India's first national art school. It was part of the Visva-Bharati University founded by Rabindranath Tagore at Shantiniketan. At Kala Bhavana, Nandalal founded the intellectual and artistic milieu to create an Indian style in art. By paying attention to the folk art forms that he saw around in Shantiniketan, he began to focus on

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Nandalal Bose, Dhaki, Haripura Posters, 1937. NGMA, New Delhi, India



K. Venkatappa, Rama's marriage, 1914. Private Collection, India

the language of art. He also illustrated primers in Bengali with woodcuts and understood the role of art in teaching new ideas. For this reason, Mahatma Gandhi invited him to paint panels that were put on display at the Congress session at Haripura in 1937. Famously called the 'Haripura Posters', they depicted ordinary rural folks busy in various activities — a musician drumming, a farmer tilling, a woman churning milk, and so on. They were painted as lively colourful sketchy figures and shown as contributing their labour to nation building. These posters echoed with Gandhi's socialist vision of including marginalised sections of Indian society through art.

Kala Bhavana, the institution where Bose taught art, inspired many young artists to carry forward this nationalist vision. It became a training ground for many artists, who taught art in different parts of the country. K. Venkatappa in South India being a prominent example. They wanted art to reach out to a wider public rather than only the elite, anglicised class of people.

Jamini Roy is a unique example of modern Indian artist, who after undergoing academic training in the colonial Art School rejected it only to adopt the flat and colourful style of folk painting seen in villages. He wanted his paintings to be simple and easy to duplicate to reach a wider public and based on themes like women and children, specifically, and rural life, generally.

However, the struggle between the Indian and European taste in art continued as seen in the art policy of the British Raj. For example, the project for mural decorations for Lutyen's Delhi buildings went to the students of Bombay School of Art, trained in realistic studies by its Principal, Gladstone Solomon. On the other hand, the Bengal School artists were allowed to decorate the Indian House in London under close British supervision.

Pan-Asianism and Modernism

The colonial art policy had created a divide between those who liked the European academic style and those who favoured Indian style. But following the Partition of Bengal in 1905, the *Swadeshi* movement was at its peak and it reflected in ideas about art. Ananda Coomaraswamy, an important art historian, wrote about *Swadeshi* in art and joined hands with a Japanese nationalist, Kakuzo Okakura, who was visiting Rabindranath Tagore in Calcutta. He came to India with his ideas about pan-Asianism, by which he wanted to unite India with other eastern nations and fight against western imperialism. Two Japanese artists accompanied him to Calcutta, who went to Shantiniketan to teach wash technique of painting to Indian students as an alternative to western oil painting.

If, on one hand, pan-Asianism was gaining popularity, ideas about modern European art also travelled to India. Hence, the year 1922 may be regarded as a remarkable one, when an important exhibition of works by Paul Klee, Kandinsky and other artists, who were part of the Bauhaus School in Germany, travelled to Calcutta. These European artists had rejected academic realism, which appealed to the *Swadeshi* artists. They created a more abstract language of art, consisting of squares, circles, lines and colour patches. For the first time, Indian artists and the public had a direct encounter with modern art of this kind. It is in the paintings by Gaganendranath Tagore, brother of Abanindranath Tagore, that the influence of modern western style of paintings can be clearly seen. He made several paintings using Cubist style, in which building interiors were created out of geometric patterns. Besides, he was deeply interested in making caricatures, in which he often made fun of rich Bengalis blindly following the European style of living.

Different Concepts of Modernism: Western and Indian

The divide between anglicists and orientalists, as mentioned earlier, was not based on race. Take the case of the Bengali intellectual, Benoy Sarkar, who sided with the anglicists and considered modernism that was growing in Europe as authentic in an article, 'The Futurism of Young Asia'. For him, the Oriental Bengal School of Art was regressive and anti-modern. On the other hand, it was E. B. Havell, an Englishman, who was in favour of return to native art to

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Amrita Sher-Gil, Camels, 1941.
NGMA, New Delhi, India

create a true modern Indian art. It is in this context that we have to view his collaboration with Abanindranath Tagore.

Amrita Sher-Gil, whom we will discuss in the next chapter, is a perfect example of the meeting of both these points of view. She used the kind of style that the Bauhaus exhibition showed to depict Indian scenes.

Modern art in India can be best understood as a result of the conflict between colonialism and nationalism. Colonialism introduced new institutions of art like art schools, exhibition galleries, art magazines and art societies. Nationalist artists, while accepting these changes, continued to assert more Indian taste in art and even accepted a larger Asian identity for a while. This legacy was going to leave a deep impact on the later history of modern Indian art. Therefore, it will keep moving from internationalism, i.e., draw ideas from the West, and indigenous, i.e., to be true to one's own legacy and tradition.

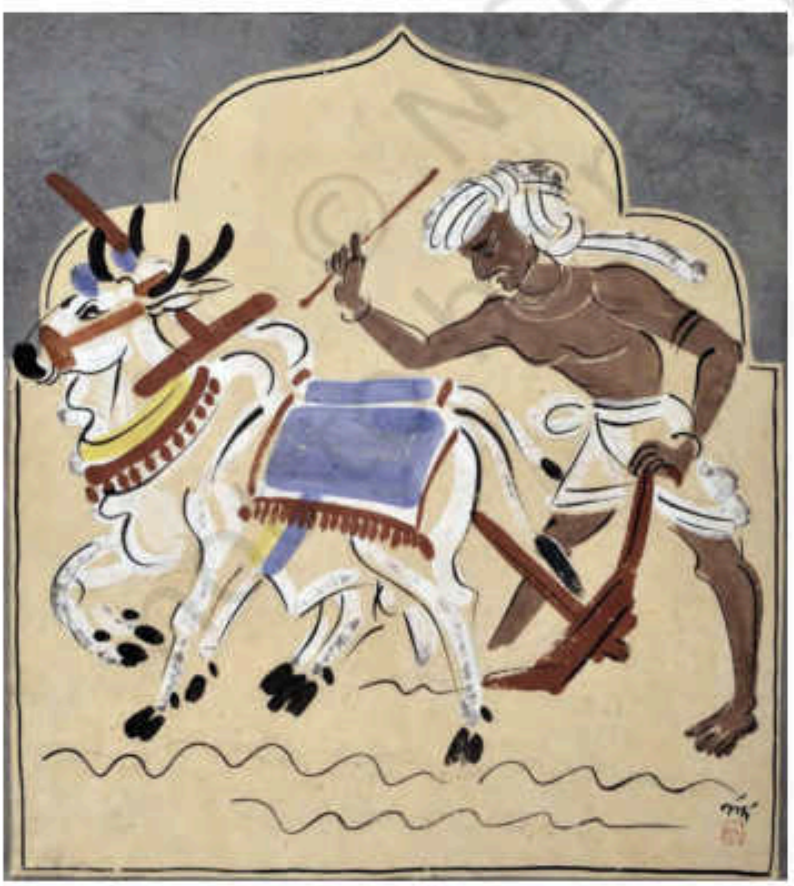
EXERCISE

1. Collect a local newspaper of the past two weeks. Select images and text from these that you consider important in the life of modern democratic state of India. With the help of these visuals and texts, compile an album that narrates the story of an independent sovereign India in the contemporary world.
2. Comment on the importance of the Bengal School artists in the making of a national style of art?
3. Write your view on any one painting by Abanindranath Tagore.
4. Which art traditions of India inspired the Bengal School artists?
5. What were the themes that Jamini Roy painted after he abandoned the academic style of painting?

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TILLER OF THE SOIL

This is one of the panels made by Nandalal Bose in 1938 for the Haripura Congress. In this panel, a farmer is shown ploughing a field— the daily activity of a common man and in a village. To capture the essence of village life in his Haripura panels, Bose made pen-and-ink brush studies of local villagers. He used thick tempera in a bold cursory style and broad brushwork. This technique and style was reminiscent of the folk art practice of *patuas* or scroll painters. Folk style is purposely used to represent the rural life. It also conveys political statement of Gandhi's idea of village. The background of the poster has an arch. The strong senses of formal design, bold colour scheme, and their blend of nature and convention in this panel reflect Bose's inspiration from Ajanta wall paintings and sculptures. More than 400 posters were prepared at Kala Bhavana under the supervision of Bose, who was influenced by the idea of Gandhi. These posters place common people in the centre of nation building. Bose utilised art to build the nation's moral character.



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The Modern Indian Art 7

Introduction to Modernism in India

Fine arts was seen as European by the British. They felt that Indians lacked training and sensibility to be able to create and appreciate fine arts. By mid and late nineteenth century, art schools were established in major cities like Lahore, Calcutta (now, Kolkata), Bombay (now, Mumbai) and Madras (now, Chennai). These art schools tended to promote traditional Indian crafts, and academic and naturalist art that reflected Victorian tastes. Even the Indian crafts, which received support, were the ones based on European taste and on the demands made by its market.

As mentioned in the previous chapter, it was against this colonial bias that nationalist art emerged, and the Bengal School of Art, as nurtured by Abanindranath Tagore and E. B. Havell, was a prime example. India's first nationalist art school, Kala Bhavana, was set up in 1919 as part of the newly established Visva-Bharati University in Shantiniketan, conceptualised by poet Rabindranath Tagore. It carried the vision of the Bengal School but also followed its own path in creating art meaningful for Indians. This was the time when the whole world was in a state of intense political turmoil in the wake of World War–I. Apart from the famous Bauhaus exhibition that travelled to Calcutta, as discussed in the previous chapter, modern European art influenced Indian artists through art magazines that were in circulation. Artists from the Tagore family—Gaganendranath and poet–painter Rabindranath, thus, knew about the international trends of Cubism and Expressionism, which had rejected academic

Gaganendranath Tagore, A Cubist City, 1925. Victoria Memorial Hall, Kolkata, India





Rabindranath Tagore, Doodle, 1920. Visva-Bharati University, Shantiniketan, West Bengal, India

realism and experimented with abstraction; They thought that art need not copy the world but create its own world out of forms, lines and colour patches. A landscape, portrait or still life may be called abstract if it draws our attention to an abstract design created by forms, lines and colour patches.

Gaganendranath Tagore used the language of Cubism to create a unique style of his own. His paintings of mysterious halls and rooms were made with vertical, horizontal and diagonal lines, which were quite different from the Cubist style of famous artist Pablo Picasso, who invented the style using geometrical facets.

Rabindranath Tagore turned to visual art quite late in life. While writing poems, he would often make patterns out of doodles and developed a unique, calligraphic style out of crossed out words. Some of these were turned into human faces and landscapes, which floated captivatingly in his poems. His palette was limited with black, yellow ochre, reds and browns. However, Rabindranath created a small visual world that was a complete departure from the more elegant and delicate style of the Bengal School, which often drew inspiration from Mughal and Pahari miniatures along with Ajanta frescoes.

Nandalal Bose in 1921–1922 joined the Kala Bhavana. His training under Abanindranath Tagore made him familiar with nationalism in art but it did not hinder him from allowing his students and other teachers to explore new avenues of artistic expression.

Benode Behari Mukherjee and Ramkinker Baij, Bose's most creative students, gave a lot of thought as how to understand the world. They developed their own unique style of sketching and painting that could capture not only their immediate environment like flora and fauna but also those who lived there. Shantiniketan had a large population of Santhal tribe at its outskirts, and these artists often painted them and made sculptures based on them. Apart from this, themes from literary sources also interested them.

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Rather than making paintings around well-known epics like *Ramayana* and *Mahabharata*, Benode Behari Mukherjee was drawn to the lives of medieval saints. On the walls of Hindi Bhavana in Shantiniketan, he made a mural called *Medieval Saints*, in which he charts a history of medieval India through the lives of Tulsi Das, Kabir and others, and focuses on their humane teachings.

Ramkinkar Baij was an artist given to the celebration of nature. His art reflects his everyday experiences. Almost all his sculptures and paintings are created as response to his environment. For instance, his *Santhal Family*, made as an outdoor sculpture within the Kala Bhavana compound, turned the daily activity of a Santhal family setting out for work into a larger than a life size piece of art. Besides, it was made out of modern material like cement mixed with pebbles, held in shape with the help of metal armature. His style was in sharp contrast with works of earlier sculptor like D. P. Roy Choudhury, who had used academic realism to celebrate the labour of working classes, *The Triumph of Labour*.



Jamini Roy, Black Horse, 1940. NGMA, New Delhi, India

If rural community was important for Benode Behari Mukherjee and Ramkinker Baij, Jamini Roy, too, made his art relevant to this context. We had briefly discussed Roy in the last chapter as an artist, who rejected his own training received at the Government School of Art, Calcutta. Being a student of Abanindranath Tagore, he realised the futility of pursuing academic art. He noticed that the rural, folk art in Bengal had much in common with how modern European masters like Picasso and Paul Klee painted. After all, Picasso had arrived at Cubism by learning from the use of bold forms found in African masks. Roy, too, used simple and pure colours. Like village artists, he also made his own colours from vegetables and minerals. His art lent itself to easy reproduction by other members in his family, quite like the artisanal practice followed in villages. However, what differentiated his art from that of village artists was that

Roy signed on his paintings. His style is seen as uniquely personal, distinct from both the academic naturalism of art schools and from Raja Ravi Varma's Indianised naturalism, as well as, from the delicate style practised by some of the Bengal School artists.

Amrita Sher-Gil (1913–1941), half Hungarian and half Indian, emerges as a unique female artist, who contributed immensely to modern Indian art through the 1930s. Unlike others, she was trained in Paris and had a first-hand experience in European modern art trends, such as Impressionism and post–Impressionism. After deciding to make India her base, she worked to develop art with Indian themes and images. Amrita Sher-Gil assimilated miniature and mural traditions of Indian art with European modernism. She died young, leaving behind a remarkable body of work, which is important for its experimental spirit and the impact it left on the next generation of Indian modernists.

Modern Ideologies and Political Art in India

Soon after Sher-Gil's death, India, still under British rule, was deeply affected by global events like World War–II. One of the indirect outcomes was the outbreak of the Bengal famine, which ravaged the region forcing massive rural migration to cities.



Prodosh Das Gupta, Twins Bronze, 1973. NGMA, New Delhi, India

The humanitarian crisis compelled many artists to reflect on their role in society. In 1943, under the leadership of Prodosh Das Gupta, a sculptor, few young artists formed the Calcutta Group, which included Nirode Mazumdar, Paritosh Sen, Gopal Ghose and Rathin Moitra. The group believed in an art that was universal in character and free from older values. They did not like the Bengal School of Art as it was too sentimental and deeply interested in the past. They wanted their paintings and sculptures to speak of their own times.

They started to simplify their visual expression by excluding details. With such an attempt, they could emphasise on elements, material, surface, forms, colours, shades and textures, etc. A comparison may be

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drawn against a sculptor from South India, P. V. Janakiram (Ganesh) who worked with metal sheets in a creative way.

Seeing abject poverty around them and the plight of people in villages and cities, many young artists in Calcutta were drawn to socialism, especially Marxism. This modern philosophy, which was taught by Karl Marx in the mid–nineteenth century in the West, asked important questions about class difference in society and appealed to these artists. They wanted their art to talk about these social problems. Chittoprasad and Somnath Hore, the two political artists of India, found printmaking to be a strong medium to express these social concerns. With printmaking, it is easier to produce multiple number of artworks and reach out to more number of people. Chittoprasad's etchings, linocuts and lithographs showed the deplorable condition of the poor. It is not surprising that he was asked by the Communist Party of India to travel to villages worst affected by the Bengal Famine and make sketches. These were later published as pamphlets under the name, *Hungry Bengal*, much to the annoyance of the British.



Chittoprasad, Hungry Bengal, 1943. Delhi Art Gallery, New Delhi, India

M. F. Husain, Farmer's Family, 1940. NGMA, New Delhi, India



The Progressive Artists' Group of Bombay and the Multifaceted Indian Art

The desire for freedom—political, as well as, artistic—soon spread widely among young artists, who witnessed Independence from the British Raj. In Bombay, another set of artists formed a group, called The Progressives in 1946. Francis Newton Souza was the outspoken leader of the group, which included M. F. Husain, K. H. Ara, S. A. Bakre, H. A. Gade and S. H. Raza. Souza wanted to question the conventions that had prevailed in art schools. For him, modern art stood for a new freedom that could challenge the traditional sense of beauty and morality. However, his experimental works were focused mainly on women, whom

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he painted as nudes, exaggerating their proportions and breaking the standard notions of beauty.

M. F. Husain, on the other hand, wanted to make the modern style of painting understandable in Indian context. For example, he would paint using the western expressionist brush strokes with bright Indian colours. He not only drew from Indian mythology and religious sources but also from the style of miniature paintings, village crafts and even folk toys.

As a result of successfully combining a modern style of painting with Indian themes, Husain's art came to eventually represent Indian modern art in the international art world. *Mother Teresa* is an example to understand how he adapted modern art to paint themes important to Indian, as well as, international audience.

Abstraction – A New Trend

While Husain largely remained a figurative artist, S.H. Raza moved in the direction of abstraction. It is not surprising then that landscape was a favourite theme for this artist. His colours ranged from bright to soft, modulated monochromes. If Husain used the figurative language of modern art to show Indian themes, Raza made a similar claim with abstraction. Some of his paintings draw from old *mandala* and *yantra* designs, and even use *bindu* as a symbol of oneness from Indian philosophy. Later, Gaitonde, too, pursued abstraction, while artists like K. K. Hebbar, S. Chavda, Akbar Padamsee, Tyeb Mehta and Krishen Khanna would keep moving between abstraction and figurative.

Abstraction was important for many sculptors like Piloo Pochkhanawala and printmakers like Krishna Reddy. For them, the use of material was as important as the new shapes they were creating. Whether in painting, printmaking or sculpture, abstraction had a wide appeal for many artists across the 1960s and 1970s. In South India, K. C. S. Paniker, who later went on to establish Cholamandalam, an artist village near Madras, was a pioneer in abstraction. In fact, he showed by imbibing artistic motifs from Tamil and Sanskrit scripts, floor decorations and rural crafts that abstraction has a long history in India.



S. H. Raza, Ma, 1972. Bombay, India

The Living Art Traditions of India 8

There has always been a timeless tradition of art forms, which have been practised for different reasons among people, living far from urban life in interior terrains of forests, deserts, mountains and villages. So far, we have studied art of a certain time, a period named after a place or dynasties, who ruled different parts of the Indian subcontinent for few hundred years or so. But what about common people? Were they not creative? Was there no art that existed around them? From where did the artists come to the courts or patrons? What did they use to make before coming to cities? Or even now, who are the unknown artists making handicrafts in faraway deserts, mountains, villages and rural areas, who have never been to an art school or design institute or even attended formal schooling?

Our country has always been a repository of indigenous knowledge, which has been transferred from one generation to another. Artists in each generation have created the best of works out of available material and technology. Many scholars named these art forms as minor arts, utility art, folk art, tribal art, people's art, ritual art, crafts, and so on. We know that these art forms have existed from time immemorial. We have seen the examples in pre-historic cave paintings or works of pottery, terracotta, bronze, ivory, etc., of the Indus period as well. During the early history and its subsequent times, we find references of artists' communities everywhere. They made pots and dresses, jewellery and ritual or votive sculptures. They decorated their walls and floors and did many more artistic things to fulfill their daily needs and supply their works to local markets at the same time. There is an instinctive aesthetic expression in their creations. There is symbolism, specific use of motifs, materials, colours and methods of making. There is a thin line between art of



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the people and crafts as both involve creativity, instinct, necessities and aesthetics.

Even now, in many pockets, we find such artifacts. In the nineteenth and twentieth centuries, a new perspective emerged among modern artists when they looked at traditional art forms around them as sources of inspiration for their creative pursuits in India, as well as, the West. In India, post-Independence a revival of handicraft industry took place. The sector became organised for commercial production. Apart from continued practice, it gained a unique identity. With the forming of States and Union Territories, each one of them showcased their unique art forms and products in their respective State emporia. The art and craft traditions of India showcase the tangible heritage of the country with history of more than five thousand years. Though we know many of these, let us talk about few of them. By and large, there has been a religious or ritualistic overtone with richer symbolism, utilitarian and decorative aspects, associated with the day-to-day practices at home to production on a large scale.

Painting Tradition

Among the many popular traditions of painting, Mithila or Madhubani painting of Bihar, Warli painting of Maharshtra, Pithoro Painting of North Gujarat and western Madhya Pradesh, Pabuji ki Phad from Rajasthan, Pichhwai of Nathdwara in Rajasthan, Gond and Sawara Paintings of Madhya Pradesh, Pata Chitra of Odisha and Bengal, etc., are few examples. Here, a few of them have been discussed.

Mithila painting

Among the most known contemporary painterly art forms is Mithila art that derives its name from Mithila, the ancient Videha and birthplace of Sita. Also called Madhubani painting after the nearest district capital, it is a widely recognised folk art tradition. It is presumed that for centuries, women living in this region have painted figures and designs on the walls of their mud houses for ceremonial occasions, particularly, weddings. People of this area see the origin of this art form at the time of Princess Sita getting married to Lord Rama.

These paintings, characterised by bright colours, are largely painted in three areas of the house— central or outer

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courtyards, eastern part of the house, which is the dwelling place of *Kuladevi*, usually, Kali, and a room in the southern part of the house, which houses the most significant images. Various armed gods and animals or images of women at work like carrying waterpots or winnowing grain, etc., are vividly portrayed in the outer central courtyard. The inner verandah, where the family shrine— *devasthana* or *gosain ghar* is located, *griha devatas* and *kula devatas* are painted. In the recent past, many paintings are done on fabric, paper, pots, etc., for commercial purposes.

The most extraordinary and colourful painting, however, is done in the part of the house known as the *kohbar ghar* or inner room, where magnificent representations of *kohbar*, a lotus with a stalk in full bloom having metaphoric and tantric connotation along with images of gods and goddesses are painted on freshly plastered walls of the room.

Among other themes that are painted are episodes from the *Bhagvata Purana*, *Ramayana*, stories of Shiva-Parvati, Durga, Kali and *Rasa-Lila* of Radha and Krishna. Mithila artists do not like empty spaces. They fill in the entire space decoratively with elements from nature like birds, flowers, animals, fish, snakes, the Sun and the moon, which often have symbolic intent, signifying love, passion, fertility, eternity, well-being and prosperity. Women paint with bamboo twigs to which some cotton swab, rice straw or fibre is attached. In earlier days, they made colour from mineral stones and organic things, such as *phalsa* and *kusum* flowers, *bilwa* leaves, *kajal*, turmeric, etc.

Warli painting

The Warli community inhabit the west coast of Northern Maharashtra around the north Sahyadri range with a large concentration in the district of Thane. Married women play a central role in creating their most important painting called *Chowk* to mark special occasions. Closely associated with the rituals of marriage, fertility, harvest and new season of sowing, *Chowk* is dominated by the figure of mother goddess, Palaghat, who is chiefly worshipped as the goddess of fertility and represents the corn goddess, Kansari.

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She is enclosed in a small square frame decorated with 'pointed' chevrons along the outer edges that symbolise Hariyali Deva, i.e., the God of Plants. Her escort and guardian is visualised as a headless warrior, riding a horse or standing beside her with five shoots of corn springing from his neck, and hence, called Panch Sirya Devata (five-headed god). He also symbolises the guardian of the fields, Khetrapal.

The central motif of Palaghat is surrounded by scenes of everyday life, portraying acts of hunting, fishing, farming, dancing, mythological stories of animals, where the tiger is conspicuously visible, scenes of buses plying and the busy urban life of Mumbai as people of Warli see around them.

These paintings are traditionally painted with rice flour on earth coloured walls of their homes. As mentioned earlier, are painted to promote fertility, these paintings avert diseases, propitiate the dead, and fulfill the demands of spirits. A bamboo stick, chewed at the end, is used as the paintbrush.

Gond painting

Gonds of Madhya Pradesh have a rich tradition with their chiefs ruling over Central India. They worshipped nature. Paintings of Gonds of Mandla and its surrounding regions have recently been transformed into a colourful depiction of animals, humans and flora. The votive paintings are geometric drawings done on the walls of huts, portraying Krishna with his cows surrounded by *gopis* with pots on their heads to which young girls and boys make offerings.





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Pithoro painting

Painted by Rathva Bhils of the Panchmahal region in Gujarat and Jhabua in the neighbouring State of Madhya Pradesh, these paintings are done on the walls of houses to mark special or thanksgiving occasions. These are large wall paintings, representing rows of numerous and magnificently coloured deities depicted as horse riders.

The rows of horse rider deities represent the cosmography of the Rathvas. The uppermost section with riders represents the world of gods, heavenly bodies and mythical creatures. An ornate wavy line separates this section from the lower region, where the wedding procession of Pithoro is depicted with minor deities, kings, goddess of destiny, an archetypal farmer, domestic animals, and so on, which represent the earth.









Pata painting

Done on fabric, palm leaf or paper, scroll painting is another example of art form practised in different parts of the country, especially, Gujarat and Rajasthan in the West and Odisha and West Bengal in East. It is also known as *Pata, Pachedi, Phad*, etc.

Bengal *patas* comprise the practice of painting on cloth (*pata*) and storytelling in regions of West Bengal. It is the most receptive oral tradition, constantly seeking new themes and formulating novel responses to major incidents in the world.

The vertically painted *pata* becomes a prop used by a *patua* (performer) for performance. *Patuas,* also called *chitrakars*, belong to communities largely settled around Midnapore, Birbhum and Bankura regions of West Bengal, parts of Bihar and Jharkhand. Handling the *pata* is their hereditary profession. They travel around villages, displaying the paintings and singing the narratives that are painted. Performances happen in common spaces of the village. The *patua* narrates three to four stories each time. After the performance, the *patua* is given alms or gift in cash or kind.

Puri *patas* or paintings evidently acquire their claim to recognition from the temple city of Puri in Odisha. It largely comprises the *pata* (initially, done on palm leaf and cloth but now done on paper as well). A range of themes are painted, such as the daily and festival *veshas* (attires) of Jagannath, Balabhadra and Subhadra (e.g., *Bada Shringar Vesha, Raghunath Vesha, Padma Vesha, Krishna–Balaram Vesha, Hariharan Vesha,* etc); *Rasa* paintings, *Ansara patti* (this substitutes the icons in the *Garbhagriha,* when they are removed for cleaning and fresh colouring is done after *Snanayatra*); *Jatri patti* (for pilgrims to take away as memorabilia and put them in personal temples at home), episodes from the myths of Jagannath, such as the *Kanchi Kaveri Pata* and *Thia-badhia pata,* a combination of aerial and lateral view of the temple with the icons and temples around or depiction of festivals around it.

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Patachitras are done on small strips of cotton cloth, which is prepared by coating the cloth with soft white stone powder and glue made from tamarind seeds. There is a practice of making the borders first. A sketch of the figures is, then, made directly with a brush and flat colours are applied. Colours, such as white, black, yellow and red are, usually, used. After completion, the painting is held over charcoal fire and lacquer is applied to the surface to make it water resistant and lend sheen to it. The colours are organic and locally procured. For example, black is obtained from lamp black, yellow and red from *haritali* and *hingal* stone, respectively, and white from powdered conch shells. Palm manuscripts are illustrated on a palm variety called *Khar-taad*. Paintings on these are not painted with brush but incised by a steel stylus, and then, filled in with ink, and sometimes, tinted with paint. There may also be some text accompanying these images. There are questions on whether to consider the palm leaf tradition a part of folk or sophisticated art as it has a lineage that stylistically connects it to the mural and palm leaf traditions of the eastern and other parts of the country.

Phads of Rajasthan

Phads are long, horizontal, cloth scrolls painted to honour folk deities of pastoral communities inhabiting the region around Bhilwara in Rajasthan. For such communities, safeguarding their livestock is the foremost concern. Such concerns purposely reflect in their myths, legends and worship patterns. Among their gods are defied cattle heroes, who are brave men who sacrificed their lives while protecting the community's cattle from robbers. Designated by the broad term *bhomia*, these heroes are honoured, worshiped and remembered for their acts of martyrdom. *Bhomias*, such as Gogaji, Jejaji, Dev Narayan, Ramdevji and Pabhuji, have inspired widespread cult following among the communities of Rabaris, Gujjars, Meghwals, Regars and others.

Illustrating the valorous tales of these *bhomias*, the *phads*, are carried by *bhopas*, the itinerant bards, who travel the territory, displaying them while narrating tales and singing devotional songs associated with these hero-deities in night-long storytelling performances. A lamp is held against the *phad* to illuminate images that are being spoken about. The *bhopa* and his companion perform to the accompaniment

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of musical instruments, such as *ravanahattha* and *veena,* and employ the *Khyal* style of singing. Through the *phads* and *phad banchan,* the community remembers the hero as a martyr and keeps his story alive.

Phads, however, are not painted by the *bhopas*. They have traditionally been painted by a caste called 'Joshis' who have been painters in the courts of the kings of Rajasthan. These painters specialised in court patronised miniature paintings. Hence, the association of skilled practitioners, bard musicians and court artists place *phads* higher than other similar cultural traditions.

Sculptural Traditions

These refer to the popular traditions of making sculptures in clay (terracotta), metal and stone. There are numerous such traditions across the country. Some of them are discussed here.

Dhokra casting

Among the popular sculptural traditions, *Dhokra* or metal sculptures made from lost wax or *cire perdue* technique is one of the most prominent metal crafts of Bastar, Chhattisgarh, parts of Madhya Pradesh, Odisha and Midnapore in West Bengal. It involves casting of bronze through the lost wax method. The metal craftsmen of Bastar are called *ghadwa*. In popular etymology, the term '*ghadwa*' means the act of shaping and creating. It is probably this that gives the casters their name. Traditionally, the *ghadwa* craftsmen, besides supplying the villagers with utensils of daily use also made jewellery, icons of locally revered deities and votive offerings in the form of snakes, elephants, horses, ritual pots, etc. Subsequently, with a decrease in demand for utensils and traditional ornaments in the community, these craftsmen began creating new (non-traditional) forms and numerous decorative objects.

Dhokra casting is an elaborate process. Black soil from the riverbank is mixed with rice husk and kneaded with water. The core figure or mould is made from this. On drying, it is covered with a second layer of cow dung mixed with clay. Resin collected from *saal* tree is, then, heated in a clay pot till it becomes liquid to which some mustard oil is also added and allowed to boil. The boiling liquid is, then, strained

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through a cloth, collected and kept in a metal vessel over water. As a result the resin solidifies but remains soft and malleable. It is, then, taken apart in small pieces, heated slightly over low burning coal and stretched into fine threads or coils. Such threads are joined together to form strips. The dried clay form is, then, overlaid with these resin strips or coils and all decorative details and eyes, nose, etc., are added to the figures. The clay form is, then, covered with layers—first of fine clay, then, with a mixture of clay and cow dung, and finally, with clay obtained from ant hills mixed with rice husk. A receptacle is, then, made from the same clay and fixed to the lower portion of the image. On the other side, a cup filled with metal pieces is sealed with the clay-rice husk mixture. For firing in the furnace, *saal* wood or its coal is preferred as fuel. The cup, containing metal, is placed in the bottom with clay moulds over it, and covered with firewood and potshards. Air is blown continuously into the furnace for about 2 to 3 hours till the metal turns into a molten state. The moulds are, then, taken out with a pair of tongs, turned upside down, given a brisk shake and the metal is poured through the receptacle. The molten metal flows exactly into the space once occupied by resin, which would by now have evaporated. The moulds are allowed to cool and the clay layer is hammered away to reveal the metal image.

Terracotta

The more ubiquitous sculptural medium prevalent across the country is terracotta. Usually, made by potters, terracotta pieces are votives or offered to local deities or used during rituals and festivals. They are made from local clay found on riverbanks or ponds. The terracotta pieces are baked for durability. Whether it is Manipur or Assam in the North-East, Kuchchha in Western India, Hills in the North, Tamil Nadu in the South, Gangetic plains or Central India, there is a variety of terracotta made by people of different regions. They are moulded, modeled by hands or made on a potter's wheel, coloured or decorated. Their forms and purposes are often similar. They are either the images of gods or goddesses. Like Ganesh, Durga or the local deity, animals, birds, insects, etc.

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Unit-II

Chapter-2

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The World Population

Distribution, Density and Growth



Not gold but only (Wo)men can make a people great and strong.

(Wo)men who for truth and honour's sake, stand fast and suffer long (Wo)men who toil while others sleep – who dare while others flee – they build a nation's pillars deep and lift it to the sky.

Ralph Waldo Emerson

The people of a country are its real wealth. It is they, who are the actual resources and make use of the country's other resources and decide its policies. Ultimately a country is known by its people.

It is important to know how many women and men a country has, how many children are born each year, how many people die and how? Whether they live in cities or villages, can they read or write and what work do they do? These are what you will study about in this unit.

The world at the beginning of 21st century recorded the presence of over 6 billion population. We shall discuss the patterns of their distribution and density here.

Why do people prefer to live in certain regions and not in others?

The population of the world is unevenly distributed. The remark of George B. Cressey about the population of Asia that "Asia has many places where people are few and few place where people are very many" is true about the pattern of population distribution of the world also.

PATTERNS OF POPULATION DISTRIBUTION IN THE WORLD

Patterns of population distribution and density help us to understand the demographic characteristics of any area. The term population distribution refers to the way people are spaced over the earth's surface. Broadly, 90 per cent of the world population lives in about 10 per cent of its land area.

The 10 most populous countries of the world contribute about 60 per cent of the world's population. Of these 10 countries, 6 are located in Asia. Identify these six countries of Asia.

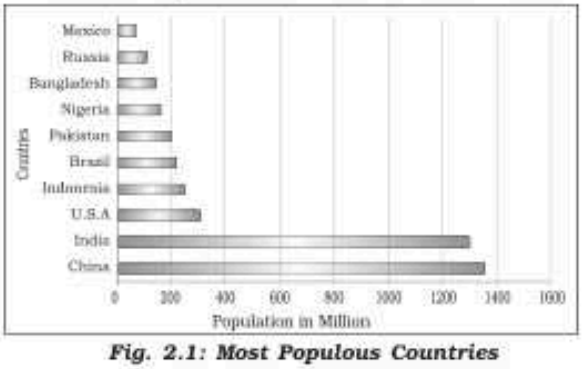


Fig. 2.1: Most Populous Countries

DENSITY OF POPULATION

Each unit of land has limited capacity to support people living on it. Hence, it is necessary to understand the ratio between the numbers of people to the size of land. This ratio is the density of population. It is usually measured in persons per sq km

$$\text{Density of Population} = \frac{\text{Population}}{\text{Area}}$$

For example, area of Region X is 100 sq km and the population is 1,50,000 persons. The density of population is calculated as:

$$\text{Density} = \frac{1,50,000}{100}$$

= 1,500 person/sq km

What does this tell you about Region X?

FACTORS INFLUENCING THE DISTRIBUTION OF POPULATION

I. Geographical Factors

(i) *Availability of water:* Water is the most important factor for life. So, people prefer to live in areas where fresh water is easily available. Water is used for drinking, bathing and cooking – and also for cattle, crops, industries and navigation. It is because of this that river valleys are among the most densely populated areas of the world.

(ii) *Landforms:* People prefer living on flat plains and gentle slopes. This is because such areas are favourable for the production of crops and to build roads and industries. The mountainous and hilly areas hinder the development of transport network and hence initially do not favour agricultural and industrial development. So, these areas tend to be less populated. The Ganga plains are among the most densely populated areas of the world while the mountains zones in the Himalayas are scarcely populated.

(iii) *Climate:* An extreme climate such as very hot or cold deserts are uncomfortable for human habitation. Areas with a comfortable climate, where there is not much seasonal variation attract more people. Areas with very heavy rainfall or extreme and harsh climates have low population. Mediterranean regions were inhabited from early periods in history due to their pleasant climate.

(iv) *Soils:* Fertile soils are important for agricultural and allied activities. Therefore, areas which have fertile loamy soils have more people living on them as these can support intensive agriculture. Can you name some areas in India which are thinly populated due to poor soils?

II. Economic Factors

(i) *Minerals:* Areas with mineral deposits attract industries. Mining and industrial activities generate employment. So, skilled and semi-skilled workers move to these areas and make them densely populated. Katanga Zambia copper belt in Africa is one such good example.

(ii) *Urbanisation:* Cities offer better employment opportunities, educational and medical facilities, better means of transport and communication. Good civic amenities and the attraction of city life draw people to the cities. It leads to rural to urban migration and cities grow in size. Mega cities of the world continue to attract large number of migrants every year.

Yet city life can be very taxing.... think of some of the unpleasant aspects of city life.

(iii) *Industrialisation:* Industrial belts provide job opportunities and attract large numbers of people. These include not just factory workers but also transport operators, shopkeepers, bank employees, doctors, teachers and other service providers. The Kobe-Osaka region of

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Japan is thickly populated because of the presence of a number of industries.

III. Social and Cultural Factors

Some places attract more people because they have religious or cultural significance. In the same way – people tend to move away from places where there is social and political unrest. Many a times governments offer incentives to people to live in sparsely populated areas or move away from overcrowded places. Can you think of some examples from your region?

POPULATION GROWTH

The population growth or population change refers to the change in number of inhabitants of a territory during a specific period of time. This change may be positive as well as negative. It can be expressed either in terms of absolute numbers or in terms of percentage. Population change in an area is an important indicator of economic development, social upliftment and historical and cultural background of the region.

Some Basic Concepts of Population Geography

Growth of Population : Change of population in particular area between two points of time is known as growth of population. For example, if we deduct the population of India 2001 (102.70 crore) from population of 2011 (121.02 crore) then we shall get the growth of population (18.15 crores) in actual numbers.

Growth Rate of Population : This is the change of population expressed in percentage.

Natural Growth of Population: This is the population increased by difference between births and deaths in a particular region between two points of time.

Natural Growth = Births – Deaths

Actual Growth of Population : This is

Births – Deaths + In Migration – Out Migration

Positive Growth of Population: This happens when the birth rate is more than the death rate between two points of time or when people from other countries migrate permanently to a region.

Negative Growth of Population: If the population decreases between two points of time it is known as negative growth of population. It occurs when the birth rate falls below the death rate or people migrate to other countries.

Components of Population Change

There are three components of population change – births, deaths and migration.

The crude birth rate (CBR) is expressed as number of live births in a year per thousand of population. It is calculated as:

$$CBR = \frac{B}{P} \times 1000$$

Here, CBR = Crude Birth Rate; B = live births during the year; P= Estimated mid year population of the area.

Death rate plays an active role in population change. Population growth occurs not only by increasing births rate but also due to decreasing death rate. Crude Death Rate (CDR) is a simple method of measuring mortality of any area. CDR is expressed in terms of number of deaths in a particular year per thousand of population in a particular region.

CDR is calculated as:

$$CDR = \frac{D}{P} \times 1000$$

Here, CDR=Crude Death Rate; D= Number of deaths; P=Estimated mid-year population of that year.

By and large mortality rates are affected by the region's demographic structure, social advancement and levels of its economic development.

Migration

Apart from birth and death there is another way by which the population size changes.

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When people move from one place to another, the place they move from is called the **Place of Origin** and the place they move to is called the **Place of Destination**. The place of origin shows a decrease in population while the population increases in the place of destination. Migration may be interpreted as a spontaneous effort to achieve a better balance between population and resources.

Migration may be permanent, temporary or seasonal. It may take place from rural to rural areas, rural to urban areas, urban to urban areas and urban to rural areas.

Do you realise that the same person is both an immigrant and an emigrant?

Immigration: Migrants who move into a new place are called Immigrants.

Emigration: Migrants who move out of a place are called Emigrants.

Can you think of reasons why people migrate?

People migrate for a better economic and social life. There are two sets of factors that influence migration.

The **Push** factors make the place of origin seem less attractive for reasons like unemployment, poor living conditions, political turmoil, unpleasant climate, natural disasters, epidemics and socio-economic backwardness.

The **Pull** factors make the place of destination seem more attractive than the place of origin for reasons like better job opportunities and living conditions, peace and stability, security of life and property and pleasant climate.

Do You Know

Human population increased more than ten times in the past 500 hundred years.

In the twentieth century itself the population has increased four times.

DEMOGRAPHIC TRANSITION

Demographic transition theory can be used to describe and predict the future population of any area. The theory tells us that population of any region changes from high births and high deaths to low births and low deaths as society progresses from rural agrarian and illiterate to urban industrial and literate society. These changes occur in stages which are collectively known as the **demographic cycle**.

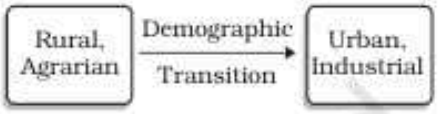


Fig. 2.2 explains the three-staged model of Demographic Transition Theory:

The first stage has high fertility and high mortality because people reproduce more to compensate for the deaths due to epidemics and variable food supply. The population growth is slow and most of the people are engaged in agriculture where large families are an asset. Life expectancy is low, people are mostly illiterate and have low levels of technology. Two hundred years ago all the countries of the world were in this stage.

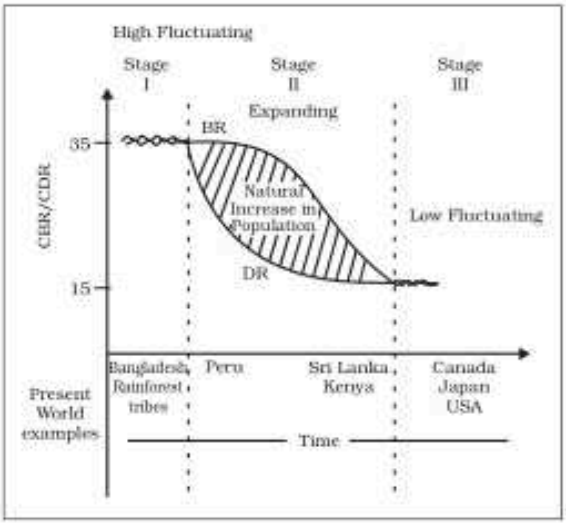


Fig. 2.2: Demographic Transition Theory

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Fertility remains high in the beginning of second stage but it declines with time. This is accompanied by reduced mortality rate. Improvements in sanitation and health conditions lead to decline in mortality. Because of this gap the net addition to population is high.

In the last stage, both fertility and mortality decline considerably. The population is either stable or grows slowly. The population becomes urbanised, literate and has high technical know- how and deliberately controls the family size.

This shows that human beings are extremely flexible and are able to adjust their fertility.

In the present day, different countries are at different stages of demographic transition.

POPULATION CONTROL MEASURES

Family planning is the spacing or preventing the birth of children. Access to family planning services is a significant factor in limiting population growth and improving women's health. Propaganda, free availability of contraceptives and tax disincentives for large families are some of the measures which can help population control.

Thomas Malthus in his theory (1798) stated that the number of people would increase faster than the food supply. Any further increase would result in a population crash caused by famine, disease and war. The preventive checks are better than the physical checks. For the sustainability of our resources, the world will have to control the rapid population increase.



EXERCISES

1. Choose the right answer from the four alternatives given below.

(i) Which one of the following continents has the highest growth of population?
- (a) Africa
- (b) South America
- (c) Asia
- (d) North America

(ii) Which one of the following is not an area of sparse population?
- (a) The Atacama
- (b) South-east Asia
- (c) Equatorial region
- (d) Polar regions

(iii) Which one of the following is not a push factor ?
- (a) Water shortage
- (b) Medical/educational facilities
- (c) Unemployment
- (d) Epidemics

(iv) Which one of the following is not a fact ?
- (a) Human population increased more than ten times during the past 500 years.
- (b) Population growth is high in the first stage of demographic transition?

2. Answer the following questions in about 30 words.

(i) Name three geographical factors that influence the distribution of population.

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Unit-III

Chapter-4

Primary Activities



Human activities which generate income are known as *economic activities*. Economic activities are broadly grouped into primary, secondary, tertiary and quaternary activities. Primary activities are directly dependent on environment as these refer to utilisation of earth's resources such as land, water, vegetation, building materials and minerals. It, thus includes, hunting and gathering, pastoral activities, fishing, forestry, agriculture, and mining and quarrying.

Why are the inhabitants of coastal and plain regions engaged in fishing and agriculture respectively? What are the physical and social factors which affect the type of primary activities in different regions?

Do You Know

People engaged in primary activities are called red-collar workers due to the outdoor nature of their work.

HUNTING AND GATHERING

The earliest human beings depended on their immediate environment for their sustenance. They subsisted on: (a) animals which they hunted; and (b) the edible plants which they gathered from forests in the vicinity.

Primitive societies depended on wild animals. People located in very cold and extremely hot climates survived on hunting. The people in the coastal areas still catch fish though fishing has experienced modernisation due to technological progress. Many species, now have become extinct or endangered due to illegal hunting (poaching). The early hunters used primitive tools made of stones, twigs or arrows so the number of animals killed was limited. Why has hunting been banned in India?

Gathering and hunting are the oldest economic activity known. These are carried out at different levels with different orientations.

Gathering is practised in regions with harsh climatic conditions. It often involves primitive societies, who extract, both plants and

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animals to satisfy their needs for food, shelter and clothing. This type of activity requires a small amount of capital investment and operates at very low level of technology. The yield per person is very low and little or no surplus is produced.



Fig. 4.1: Women Gathering Oranges in Mizoram

Gathering is practised in: (i) high latitude zones which include northern Canada, northern Eurasia and southern Chile; (ii) Low latitude zones such as the Amazon Basin, tropical Africa, Northern fringe of Australia and the interior parts of Southeast Asia (Fig. 4.2).

In modern times some gathering is market-oriented and has become commercial. Gatherers collect valuable plants such as leaves, barks of trees and medicinal plants and after simple processing sell the products in the market. They use various parts of the plants, for example, the bark is used for quinine, tanin extract and cork— leaves supply materials for beverages, drugs, cosmetics, fibres, thatch and fabrics; nuts for food and oils and tree trunk yield rubber, balata, gums and resins.

Do You Know

The name of the part of the chewing gum after the flavour is gone? It is called *Chicle* — it is made from the milky juice of zapota tree.

Gathering has little chance of becoming important at the global level. Products of such an

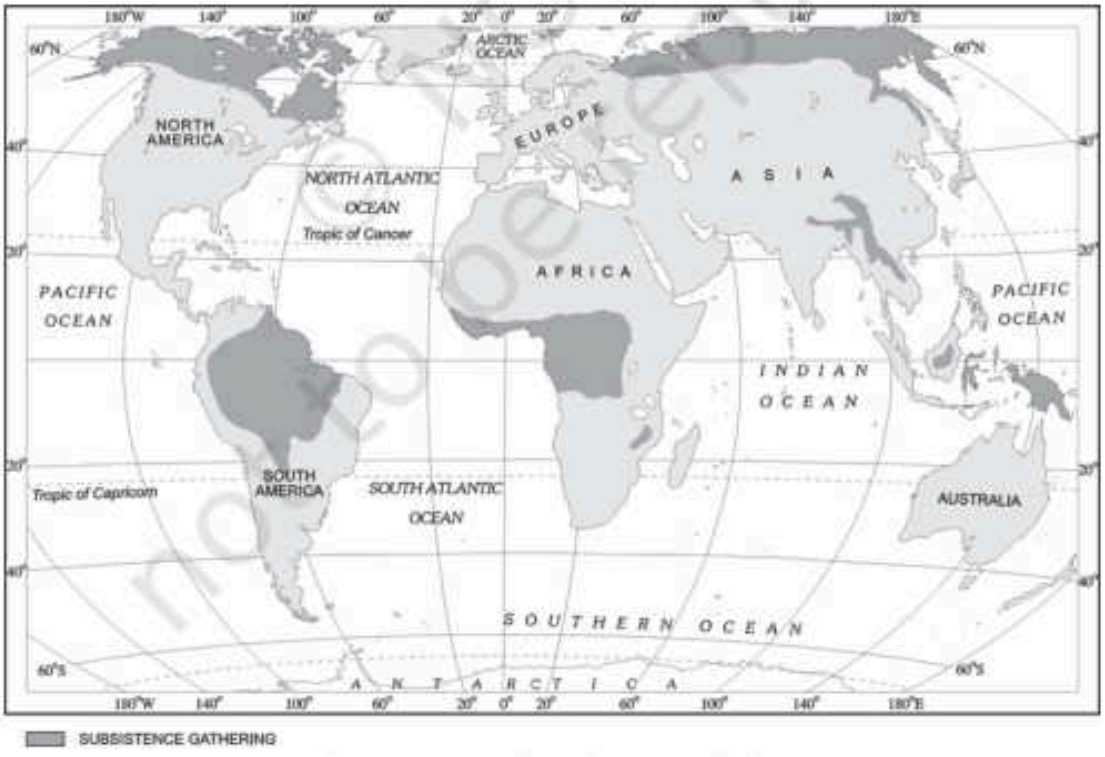


Fig. 4.2: Areas of Subsistence Gathering

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activity cannot compete in the world market. Moreover, synthetic products often of better quality and at lower prices, have replaced many items supplied by the gatherers in tropical forests.

PASTORALISM

At some stage in history, with the realisation that hunting is an unsustainable activity, human beings might have thought of domestication of animals. People living in different climatic conditions selected and domesticated animals found in those regions. Depending on the geographical factors, and technological development, animal rearing today is practised either at the subsistence or at the commercial level.

Nomadic Herding

Nomadic herding or pastoral nomadism is a primitive subsistence activity, in which the herders rely on animals for food, clothing, shelter, tools and transport. They move from one place to another along with their livestock, depending on the amount and quality of pastures and water. Each nomadic community occupies a well-identified territory as a matter of tradition.



Fig. 4.3: Nomads taking their sheep up to the Mountains at the onset of summer

A wide variety of animals is kept in different regions. In tropical Africa, cattle are the most important livestock, while in Sahara and Asiatic deserts, sheep, goats and camel are reared. In the mountainous areas of Tibet and Andes, yak and llamas and in the Arctic and sub Arctic areas, reindeer are the most important animals.

Pastoral nomadism is associated with three important regions. The core region extends from the Atlantic shores of North Africa eastwards across the Arabian peninsula into Mongolia and Central China. The second region extends over the tundra region of Eurasia. In the southern hemisphere there are small areas in South-west Africa and on the island of Madagascar (Fig. 4.4)

Movement in search of pastures is undertaken either over vast horizontal distances or vertically from one elevation to another in the mountainous regions. The process of migration from plain areas to pastures on mountains during summers and again from mountain pastures to plain areas during winters is known as *transhumance*. In mountain regions, such as Himalayas, Gujjars, Bakarwals, Gaddis and Bhotiyas migrate from plains to the mountains in summers and to the plains from the high altitude pastures in winters. Similarly, in the tundra regions, the nomadic herders move from south to north in summers and from north to south in winters.

The number of pastoral nomads has been decreasing and the areas operated by them shrinking. This is due to (a) imposition of political boundaries; (b) new settlement plans by different countries.

Commercial Livestock Rearing

Unlike nomadic herding, commercial livestock rearing is more organised and capital intensive. Commercial livestock ranching is essentially associated with western cultures and is practised on permanent ranches. These ranches cover large areas and are divided into a number of parcels, which are fenced to regulate the grazing. When the grass of one parcel is grazed, animals are moved to another parcel. The number of animals in a pasture is kept according to the carrying capacity of the pasture.

This is a specialised activity in which only one type of animal is reared. Important animals include sheep, cattle, goats and horses. Products such as meat, wool, hides and skin are processed and packed scientifically and exported to different world markets.

Rearing of animals in ranching is organised on a scientific basis. The main

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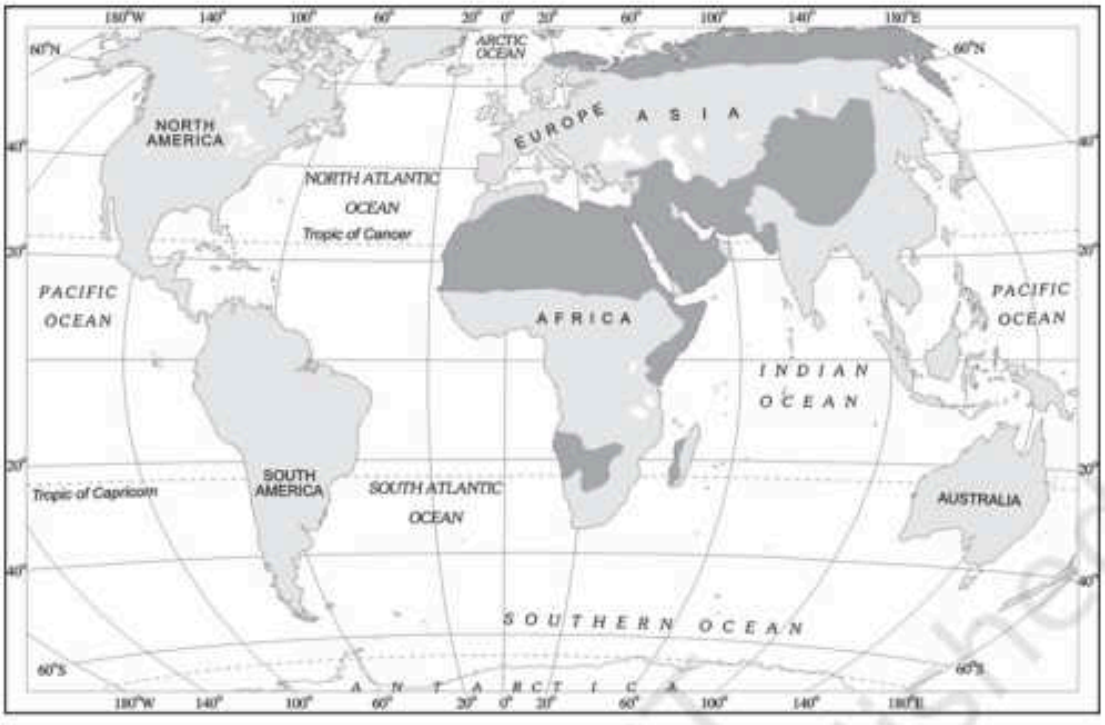


Fig. 4.4: Areas of Nomadic Herding



Fig. 4.5: Commercial Livestock Rearing

Reindeer rearing in the northern regions of Alaska where most of the Eskimos own about two-third of the stock.

emphasis is on breeding, genetic improvement, disease control and health care of the animals.

New Zealand, Australia, Argentina, Uruguay and United States of America are important countries where commercial livestock rearing is practised (Fig. 4.6).

AGRICULTURE

Agriculture is practised under multiple combinations of physical and socio-economic conditions, which gives rise to different types of agricultural systems.

Based on methods of farming, different types of crops are grown and livestock raised. The following are the main agricultural systems.

Subsistence Agriculture

Subsistence agriculture is one in which the farming areas consume all, or nearly so, of the products locally grown. It can be grouped in two categories — Primitive Subsistence Agriculture and Intensive Subsistence Agriculture.

Primitive Subsistence Agriculture

Primitive subsistence agriculture or shifting cultivation is widely practised by many tribes in the tropics, especially in Africa, south and central America and south east Asia (Fig. 4.7).

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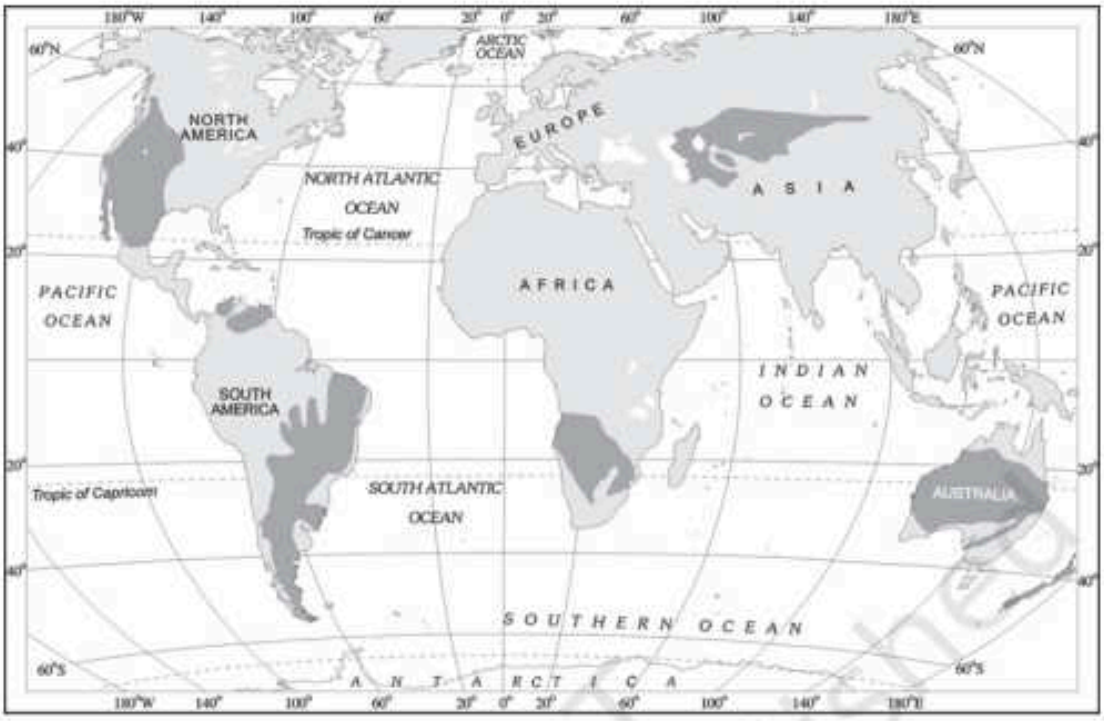


Fig. 4.6: Areas of Commercial Livestock Rearing

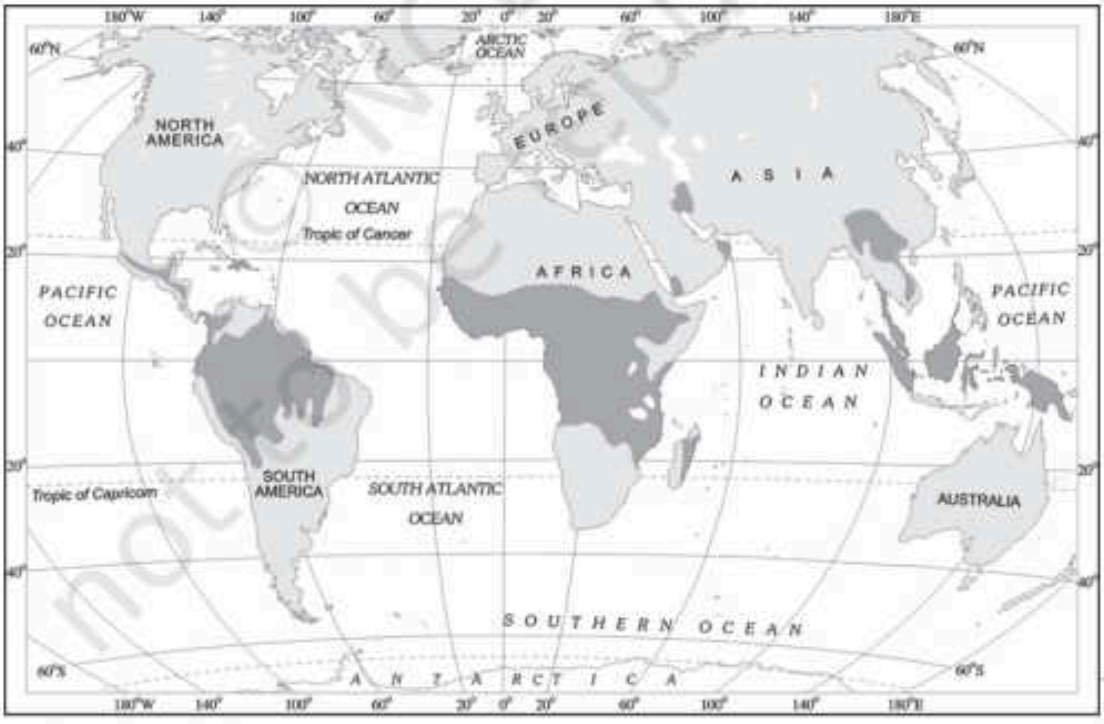


Fig. 4.7: Areas of Primitive Subsistence Agriculture

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The vegetation is usually cleared by fire, and the ashes add to the fertility of the soil. Shifting cultivation is thus, also called **slash and burn agriculture**. The cultivated patches are very small and cultivation is done with very primitive tools such as sticks and hoes. After sometime (3 to 5 years) the soil looses its fertility and the farmer shifts to another parts and clears other patch of the forest for cultivation. The farmer may return to the earlier patch after sometime. One of the major problems of shifting cultivation is that the cycle of *jhum* becomes less and less due to loss of fertility in different parcels. It is prevalent in tropical region in different names, e.g. **Jhuming** in North eastern states of India, **Milpa** in central America and Mexico and **Ladang** in Indonesia and Malaysia. Find out other areas and the names with which shifting cultivation is done.

Intensive Subsistence Agriculture

This type of agriculture is largely found in densely populated regions of monsoon Asia.

Basically, there are two types of intensive subsistence agriculture.

(i) *Intensive subsistence agriculture dominated by wet paddy cultivation:* This type of agriculture is characterised by dominance of the rice crop. Land holdings are very small due to the high density of population. Farmers work with the help of family labour leading to intensive use of land. Use of machinery is limited and most of the agricultural operations are done by manual labour. Farm yard manure is used to maintain the fertility of the soil. In this type of agriculture, the yield per unit area is high but per labour productivity is low.

(ii) *Intensive subsistence agriculture dominated by crops other than paddy:* Due to the difference in relief, climate, soil and some of the other geographical factors, it is not practical to grow paddy in many parts of monsoon Asia. Wheat, soyabean, barley and sorghum are grown in northern China, Manchuria, North Korea and North Japan. In India wheat is grown in western

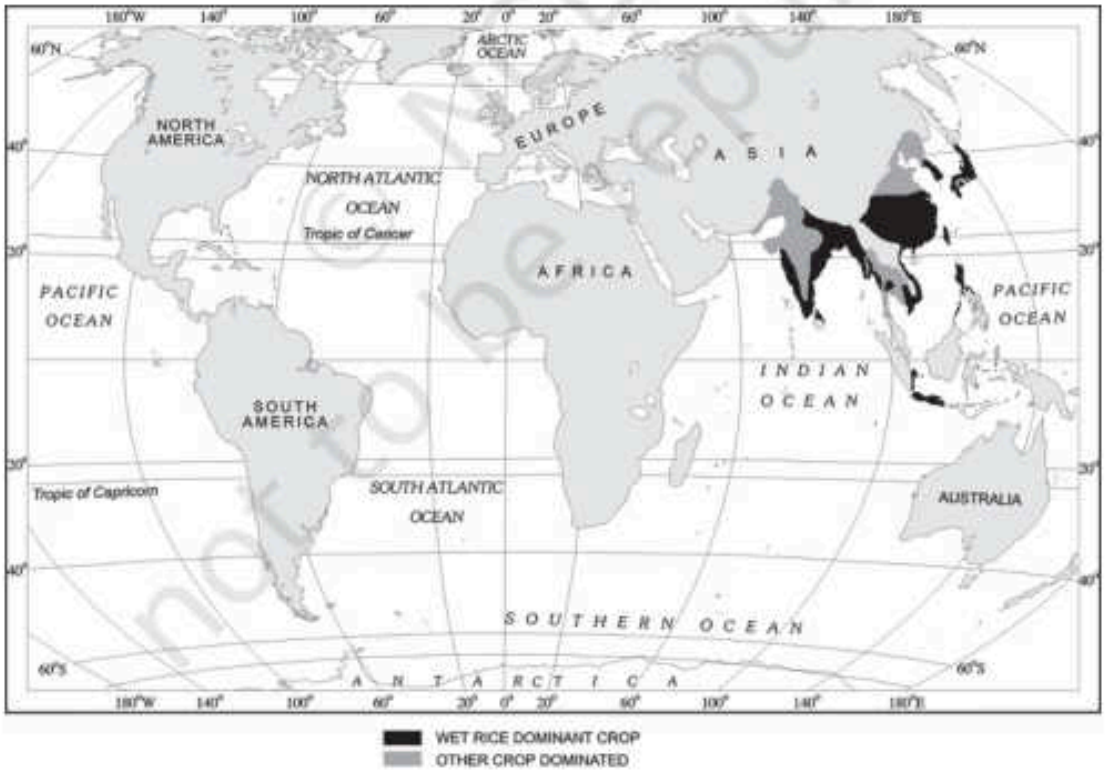


Fig. 4.8: Areas of Intensive Subsistence Farming



Fig. 4.9: Rice Transplantation

parts of the Indo-Gangetic plains and millets are grown in dry parts of western and southern India. Most of the characteristics of this type of agriculture are similar to those dominated by wet paddy except that irrigation is often used.

The Europeans colonised many parts in the world and they introduced some other forms of agriculture such as plantations which were mainly profit-oriented large scale production systems.

Plantation Agriculture

Plantation agriculture as mentioned above was introduced by the Europeans in colonies situated in the tropics. Some of the important plantation crops are tea, coffee, cocoa, rubber, cotton, oil palm, sugarcane, bananas and pineapples.

The characteristic features of this type of farming are large estates or plantations, large capital investment, managerial and technical support, scientific methods of cultivation, single crop specialisation, cheap labour, and a good system of transportation which links the estates to the factories and markets for the export of the products.

The French established cocoa and coffee plantations in west Africa. The British set up large tea gardens in India and Sri Lanka, rubber plantations in Malaysia and sugarcane and banana plantations in West Indies. Spanish and Americans invested heavily in coconut and sugarcane plantations in the Philippines. The Dutch once had monopoly over sugarcane plantation in Indonesia. Some coffee fazendas (large plantations) in Brazil are still managed by Europeans.

Today, ownership of the majority of plantations has passed into the hands of the government or the nationals of the countries concerned.



Fig. 4.10: Tea Plantation

The slopes of hills are used for tea plantations because of favourable geographical conditions.

Extensive Commercial Grain Cultivation

Commercial grain cultivation is practised in the interior parts of semi-arid lands of the mid-latitudes. Wheat is the principal crop, though other crops like corn, barley, oats and rye are also grown. The size of the farm is very large, therefore entire operations of cultivation from ploughing to harvesting are mechanised (Fig. 4.11). There is low yield per acre but high yield per person. Why does this happen?

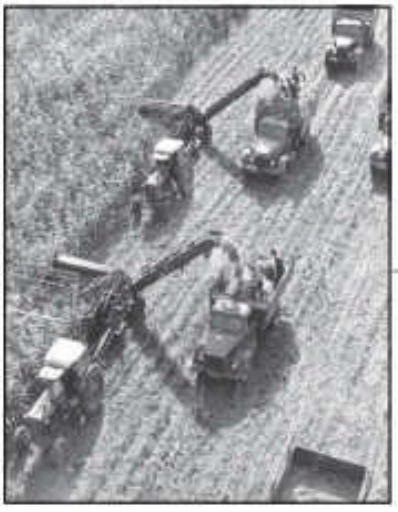


Fig. 4.11: Mechanised Grain Farming

Combine crews are capable of harvesting grain over many hectares in a single day.

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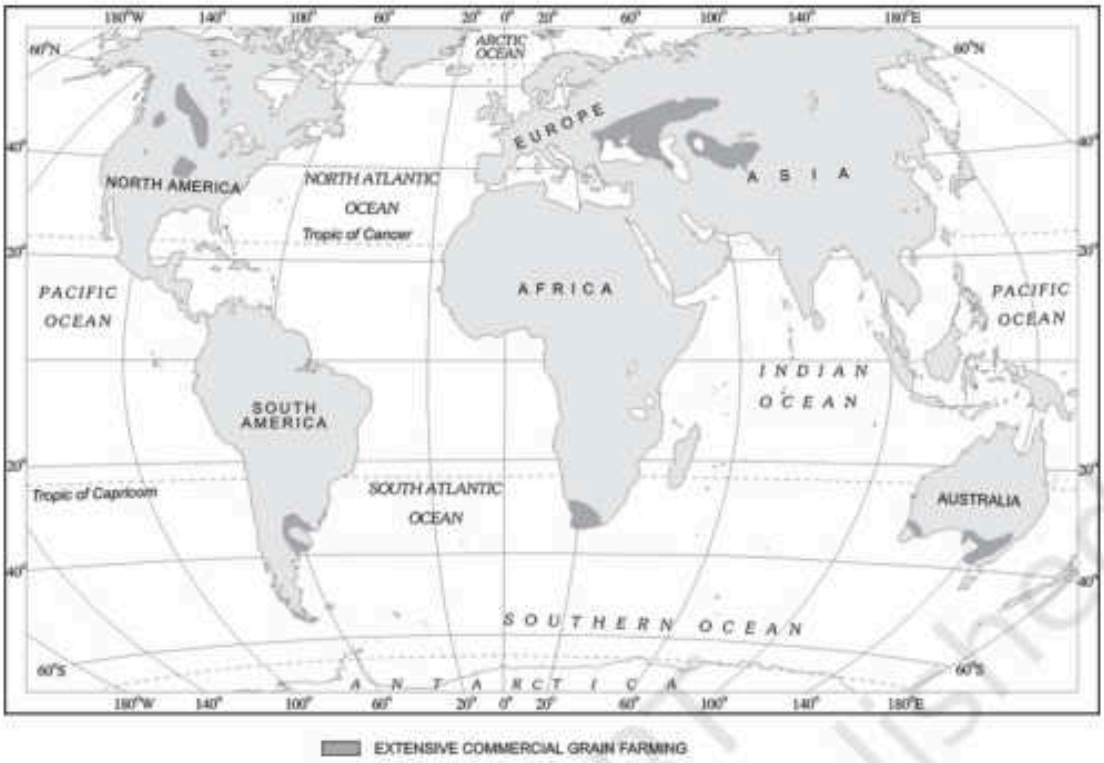


Fig. 4.12: Areas of Extensive Commercial Grain Farming

This type of agriculture is best developed in Eurasian steppes, the Canadian and American Prairies, the Pampas of Argentina, the Velds of South Africa, the Australian Downs and the Canterbury Plains of New Zealand. (Locate these areas on the world map).

Mixed Farming

This form of agriculture is found in the highly developed parts of the world, e.g. North-western Europe, Eastern North America, parts of Eurasia and the temperate latitudes of Southern continents (Fig. 4.14).

Mixed farms are moderate in size and usually the crops associated with it are wheat, barley, oats, rye, maize, fodder and root crops. Fodder crops are an important component of mixed farming. Crop rotation and intercropping play an important role in maintaining soil fertility. Equal emphasis is laid on crop cultivation and animal husbandry. Animals like cattle, sheep, pigs and poultry provide the main income along with crops.

Mixed farming is characterised by high capital expenditure on farm machinery and building, extensive use of chemical fertilisers and green manures and also by the skill and expertise of the farmers.

Dairy Farming

Dairy is the most advanced and efficient type of rearing of milch animals. It is highly capital intensive. Animal sheds, storage facilities for fodder, feeding and milching machines add to the cost of dairy farming. Special emphasis is laid on cattle breeding, health care and veterinary services.



Fig. 4.13: A Dairy Farm in Austria

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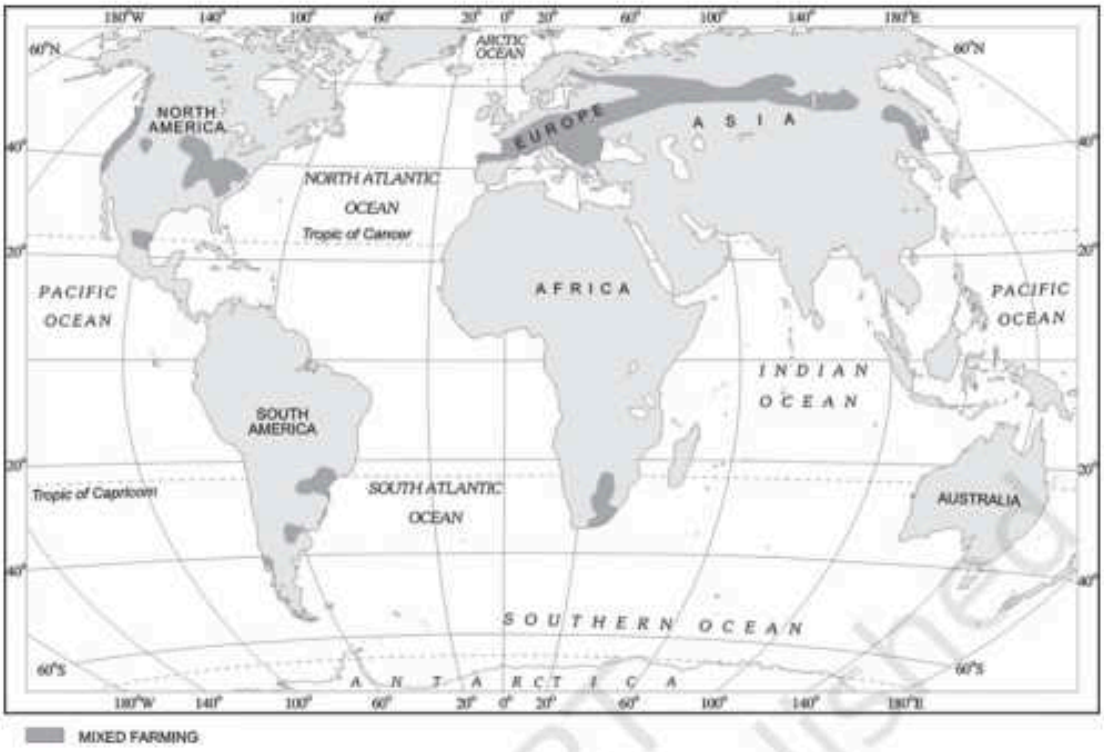


Fig. 4.14: Areas of Mixed Farming

It is highly labour intensive as it involves rigorous care in feeding and milching. There is no off season during the year as in the case of crop raising.

It is practised mainly near urban and industrial centres which provide neighbourhood market for fresh milk and dairy products. The development of transportation, refrigeration, pasteurisation and other preservation processes have increased the duration of storage of various dairy products.

There are three main regions of commercial dairy farming. The largest is North Western Europe the second is Canada and the third belt includes South Eastern Australia, New Zealand and Tasmania (Fig. 4.16).

Mediterranean Agriculture

Mediterranean agriculture is highly specialised commercial agriculture. It is practised in the countries on either side of the Mediterranean



Fig. 4.15 (a): A vineyard in Switzerland



Fig. 4.15 (b): Collection of grapes in a collective farm of Kazakhstan

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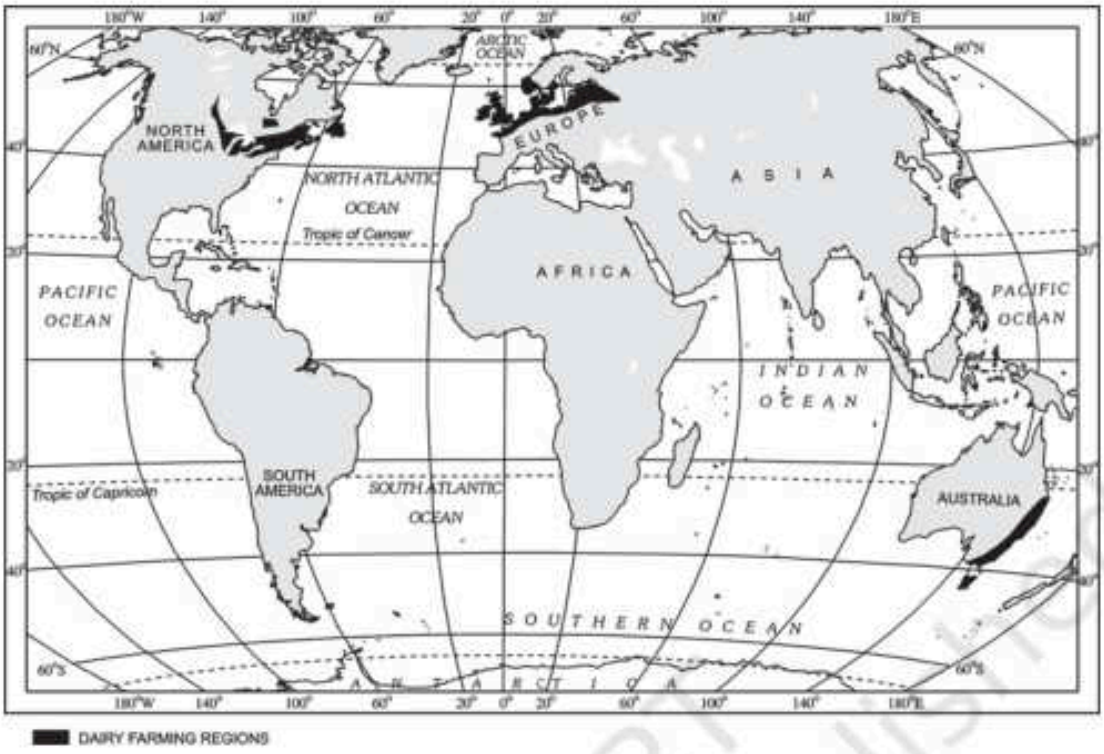


Fig. 4.16: Areas of Dairy Farming

sea in Europe and in north Africa from Tunisia to Atlantic coast, southern California, central Chile, south western parts of South Africa and south and south western parts of Australia. This region is an important supplier of citrus fruits.

Viticulture or grape cultivation is a speciality of the Mediterranean region. Best quality wines in the world with distinctive flavours are produced from high quality grapes in various countries of this region. The inferior grapes are dried into raisins and currants. This region also produces olives and figs. The advantage of Mediterranean agriculture is that more valuable crops such as fruits and vegetables are grown in winters when there is great demand in European and North American markets.

Market Gardening and Horticulture

Market gardening and horticulture specialise in the cultivation of high value crops such as vegetables, fruits and flowers, solely for the urban markets. Farms are small and are located where there are good transportation links with the urban centre where high income group of consumers is located. It is both labour and capital intensive and lays emphasis on the use of irrigation, HYV seeds, fertilisers, insecticides, greenhouses and artificial heating in colder regions.

This type of agriculture is well developed in densely populated industrial districts of north west Europe, north eastern United States of America and the Mediterranean regions. The Netherlands specialises in growing flowers and horticultural crops especially tulips, which are flown to all major cities of Europe.

The regions where farmers specialise in vegetables only, the farming is know as **truck farming**. The distance of truck farms from the market is governed by the distance that a truck can cover overnight, hence the name truck farming.

In addition to market gardening, a modern development in the industrial regions of Western Europe and North America is factory farming. Livestock, particularly poultry and cattle

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Figure 4.17 (a): Vegetables being grown in the vicinity of the city



Figure 4.17 (b): Vegetables being loaded into a truck and cycle carts for transporting to city markets

rearing, is done in stalls and pens, fed on manufactured feedstuff and carefully supervised against diseases. This requires heavy capital investment in terms of building, machinery for various operations, veterinary services and heating and lighting. One of the important features of poultry farming and cattle rearing is breed selection and scientific breeding.

Types of farming can also be categorised according to the farming organisation. Farming organisation is affected by the way in which farmers own their farms and various policies of the government which help to run these farms.

Co-operative Farming

A group of farmers form a co-operative society by pooling in their resources voluntarily for more efficient and profitable farming. Individual farms remain intact and farming is a matter of cooperative initiative.

Co-operative societies help farmers, to procure all important inputs of farming, sell the products at the most favourable terms and help in processing of quality products at cheaper rates.

Co-operative movement originated over a century ago and has been successful in many western European countries like Denmark, Netherlands, Belgium, Sweden, Italy etc. In Denmark, the movement has been so successful that practically every farmer is a member of a co-operative.

Collective Farming

The basic principle behind this types of farming is based on social ownership of the means of production and collective labour. Collective farming or the model of **Kolkhoz** was introduced in erstwhile Soviet Union to improve upon the inefficiency of the previous methods of agriculture and to boost agricultural production for self-sufficiency.

The farmers used to pool in all their resources like land, livestock and labour. However, they were allowed to retain very small plots to grow crops in order to meet their daily requirements.

MINING

The discovery of minerals in the history of human development, is reflected in many stages in terms of copper age, bronze age and iron age. The use of minerals in ancient times was largely confined to the making of tools, utensils and weapons. The actual development of mining began with the industrial revolution and its importance is continuously increasing.

Factors Affecting Mining Activity

The profitability of mining operations thus, depends on two main factors:

(i) Physical factors include the size, grade and the mode of occurrence of the deposits.

(ii) Economic factors such as the demand for the mineral, technology available and used, capital to develop infrastructure and the labour and transport costs.

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Fig. 4.18: Oil drilling operation in the Gulf of Mexico

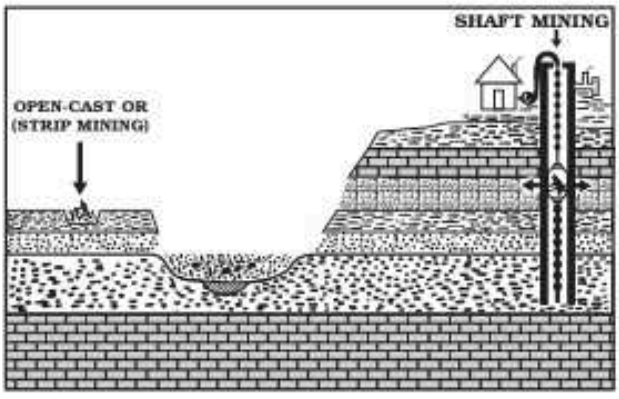


Fig. 4.19: Methods of Mining

Methods of Mining

Depending on the mode of occurrence and the nature of the ore, mining is of two types: surface and underground mining. The surface mining also known as *open-cast* mining is the easiest and the cheapest way of mining minerals that occur close to the surface. Overhead costs such as safety precautions and equipment is relatively low in this method. The output is both large and rapid.

When the ore lies deep below the surface, **underground mining method** (shaft method) has to be used. In this method, vertical shafts have to be sunk, from where underground galleries radiate to reach the minerals. Minerals are extracted and transported to the surface through these passages. It requires specially designed lifts, drills, haulage vehicles, ventilation system for safety and efficient movement of people and material. This method is risky. Poisonous gases, fires, floods and caving in lead to fatal accidents. Have you ever read about mine fires and flooding of coal mines in India?

The developed economies are retreating from mining, processing and refining stages of production due to high labour costs, while the developing countries with large labour force and striving for higher standard of living are becoming more important. Several countries of Africa and few of south America and Asia have over fifty per cent of the earnings from minerals alone.

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Unit-III

Chapter-5

Secondary Activities



All economic activities namely primary, secondary, tertiary and quaternary, revolve around obtaining and utilising resources necessary for survival.

Secondary activities add value to natural resources by *transforming* raw materials into valuable products. Cotton in the boll has limited use but after it is transformed into yarn, becomes more valuable and can be used for making clothes. Iron ore, cannot be used; directly from the mines, but after being converted into steel it gets its value and can be used for making many valuable machines, tools, etc. The same is true of most of the materials from the farm, forest, mine and the sea. Secondary activities, therefore, are concerned with manufacturing, processing and construction (infrastructure) industries.

MANUFACTURING

Manufacturing involves a full array of production from handicrafts to moulding iron and steel and stamping out plastic toys to assembling delicate computer components or space vehicles. In each of these processes, the common characteristics are the application of power, mass production of identical products and specialised labour in factory settings for the production of standardised commodities. Manufacturing may be done with modern power and machinery or it may still be very primitive. Most of the Third World countries still 'manufacture' in the literal sense of the term. It is difficult to present a full picture of all the manufacturers in these countries. More emphasis is given to the kind of 'industrial' activity which involves less complicated systems of production.

Characteristics of Modern Large Scale Manufacturing

Modern large scale manufacturing has the following characteristics:

Specialisation of Skills/Methods of Production

Under the 'craft' method factories produce only a few pieces which are made-to-order. So the costs are high. On the other hand, mass

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production involves production of large quantities of standardised parts by each worker performing only one task repeatedly.

'Manufacturing' Industry and 'Manufacturing Industry'

Manufacturing literally means *'to make by hand'*. However, now it includes goods 'made by machines'. It is essentially a process which involves *transforming raw materials into finished goods of higher value for sale in local or distant markets.* Conceptually, an industry is a geographically located manufacturing unit maintaining books of accounts and, records under a management system. As the term *industry* is comprehensive, it is also used as synonymous with 'manufacturing' When one uses terms like 'steel industry' and 'chemical industry' one thinks of *factories* and *processes.* But there are many secondary activities which are not carried on in factories such as what is now called the 'entertainment industry' and Tourism industry, etc. So for clarity the longer expression 'manufacturing industry' is used.

Mechanisation

Mechanisation refers to using gadgets which accomplish tasks. Automation (without aid of human thinking during the manufacturing process) is the advanced stage of mechanisation. Automatic factories with feedback and closed-loop computer control systems where machines are developed to 'think', have sprung up all over the world.

Technological Innovation

Technological innovations through research and development strategy are an important aspect of modern manufacturing for quality control, eliminating waste and inefficiency, and combating pollution.

Organisational Structure and Stratification

Modern manufacturing is characterised by a complex machine technology and extreme specialisation and division of labour for producing more goods with less efforts and low costs. It also requires vast capital, large organisations and executive bureaucracy for a large scale manufacturing industry.

Uneven Geographic Distribution

Major concentrations of modern manufacturing have flourished in a few number of places. These cover less than 10 per cent of the world's land area. These nations have become the centres of economic and political power. However, in terms of the total area covered, manufacturing sites are much less conspicuous and concentrated on much smaller areas than that of agriculture due to greater intensity of processes. For example, 2.5 sq km of the American corn belt usually includes about four large farms employing about 10-20 workers supporting 50-100 persons. But this same area could contain several large integrated factories and employ thousands of workers.

Why do Large-scale Industries choose different locations?

Industries maximise profits by reducing costs. Therefore, industries should be located at points where the production costs are minimum. Some of the factors influencing industrial locations are as under:

Access to Market

The existence of a market for manufactured goods is the most important factor in the location of industries. 'Market' means people who have a demand for these goods and also have the purchasing power (ability to purchase) to be able to purchase from the sellers at a place. Remote areas inhabited by a few people offer small markets. The developed regions of Europe, North America, Japan and Australia provide large global markets as the purchasing power of the people is very high. The densely populated regions of South and South-east Asia also

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provide large markets. Some industries, such as aircraft manufacturing, have a global market. The arms industry also has global markets.

Access to Raw Material

Raw material used by industries should be cheap and easy to transport. Industries based on cheap, bulky and weight-losing material (ores) are located close to the sources of raw material such as steel, sugar, and cement industries. Perishability is a vital factor for the industry to be located closer to the source of the raw material. Agro-processing and dairy products are processed close to the sources of farm produce or milk supply respectively.

Access to Labour Supply

Labour supply is an important factor in the location of industries. Some types of manufacturing still require skilled labour. Increasing mechanisation, automation and flexibility of industrial processes have reduced the dependence of industry upon the labours.

Access to Sources of Energy

Industries which use more power are located close to the source of the energy supply such as the aluminium industry.

Earlier coal was the main source of energy, today hydroelectricity and petroleum are also important sources of energy for many industries.

Access to Transportation and Communication Facilities

Speedy and efficient transport facilities to carry raw materials to the factory and to move finished goods to the market are essential for the development of industries. The cost of transport plays an important role in the location of industrial units. Western Europe and eastern North America have a highly developed transport system which has always induced the concentration of industries in these areas. Modern industry is inseparably tied to transportation systems. Improvements in transportation led to integrated economic development and regional specialisation of manufacturing.

Communication is also an important need for industries for the exchange and management of information.

Government Policy

Governments adopt 'regional policies' to promote 'balanced' economic development and hence set up industries in particular areas.

Access to Agglomeration Economies/ Links between Industries

Many industries benefit from nearness to a leader-industry and other industries. These benefits are termed as agglomeration economies. Savings are derived from the linkages which exist between different industries.

These factors operate together to determine industrial location.

Foot Loose Industries

Foot loose industries can be located in a wide variety of places. They are not dependent on any specific raw material, weight losing or otherwise. They largely depend on component parts which can be obtained anywhere. They produce in small quantity and also employ a small labour force. These are generally not polluting industries. The important factor in their location is accessibility by road network.

Classification of Manufacturing Industries

Manufacturing industries are classified on the basis of their size, inputs/raw materials, output/products and ownership (Fig. 5.1).

Industries based on Size

The amount of capital invested, number of workers employed and volume of production determine the size of industry. Accordingly, industries may be classified into household or cottage, small-scale and large-scale.

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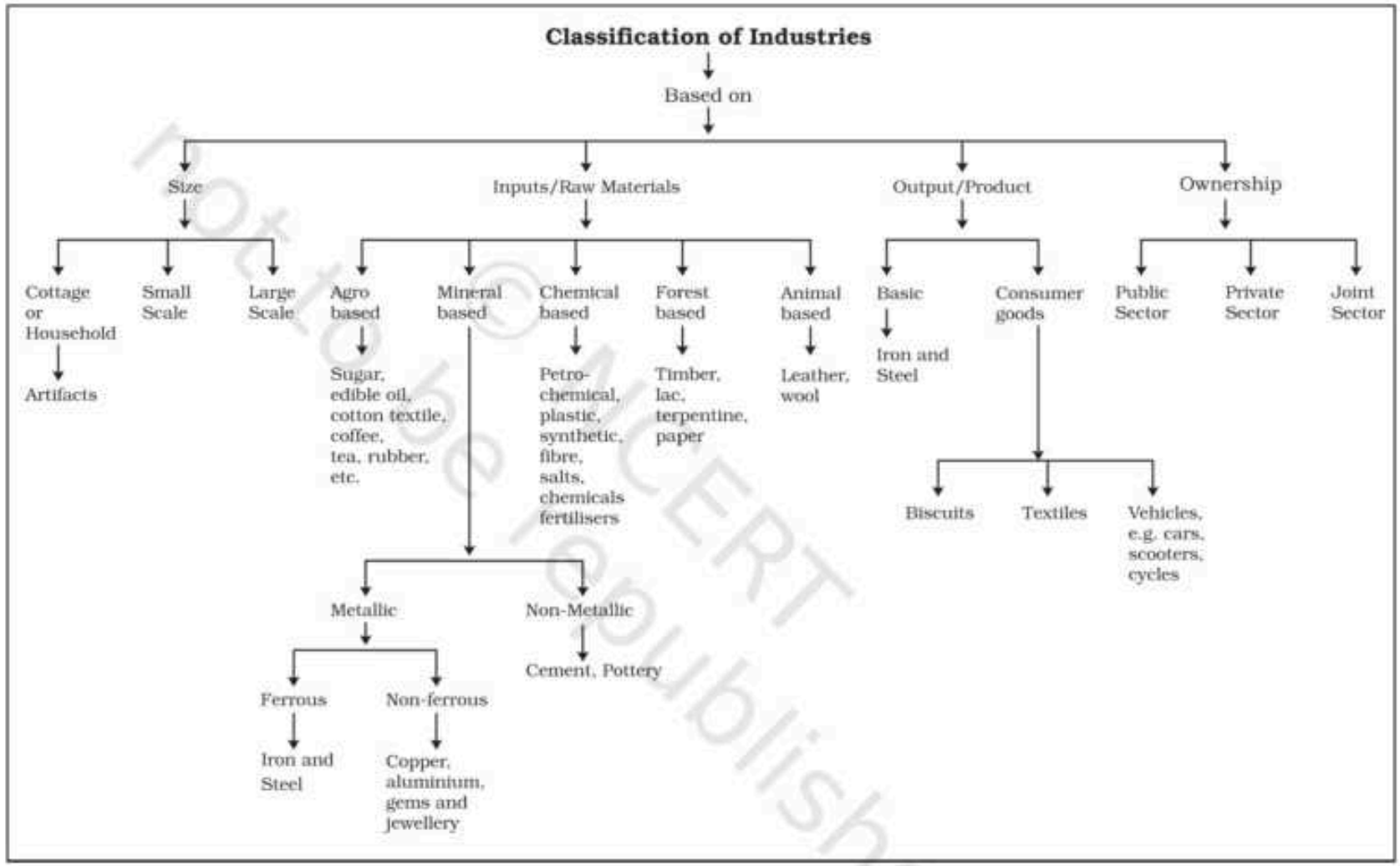


Fig. 5.1 : Classification of Industries

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HOUSEHOLD INDUSTRIES OR COTTAGE MANUFACTURING

It is the smallest manufacturing unit. The artisans use local raw materials and simple tools to produce everyday goods in their homes with the help of their family members or part-time labour. Finished products may be for consumption in the same household or, for sale in local (village) markets, or, for barter. Capital and transportation do not wield much influence as this type of manufacturing has low commercial significance and most of the tools are devised locally.



Fig. 5.2 (a) : A man making pots in his courtyard-example of household industry in Nagaland



Fig. 5.2 (b) : A man weaving a bamboo basket by the roadside in Arunachal Pradesh

Some common everyday products produced in this sector of manufacturing include foodstuffs, fabrics, mats, containers, tools, furniture, shoes, and figurines from wood lot and forest, shoes, thongs and other articles from leather; pottery and bricks from clays and stones. Goldsmiths make jewellery of gold, silver and bronze. Some artefacts and crafts are made out of bamboo, wood obtained locally from the forests.

Small Scale Manufacturing

Small scale manufacturing is distinguished from household industries by its production techniques and place of manufacture (a workshop outside the home/cottage of the producer). This type of manufacturing uses local raw material, simple power-driven machines and semi-skilled labour. It provides employment and raises local purchasing power. Therefore, countries like India, China, Indonesia and Brazil, etc. have developed labour-intensive small scale manufacturing in order to provide employment to their population.



Fig. 5.3: Products of cottage industry on sale in Assam

Large Scale Manufacturing

Large scale manufacturing involves a large market, various raw materials, enormous energy, specialised workers, advanced technology, assembly-line mass production and large capital. This kind of manufacturing developed in the last 200 years, in the United Kingdom, north-eastern U.S.A. and Europe. Now it has diffused to almost all over the world.

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On the basis of the system of large scale manufacturing, the world's major industrial regions may be grouped under two broad types, namely

(i) traditional large-scale industrial regions which are thickly clustered in a few more developed countries.

(ii) high-technology large scale industrial regions which have diffused to less developed countries.



Fig. 5.4 : Passenger car assembly hires at a plant of the Motor Company in Japan



Fig. 5.5: Tea Garden and a Tea Factory in the Nilgiri Hills of Tamil Nadu

Agri-business is commercial farming on an industrial scale often financed by business whose main interests lie outside agriculture, for example, large corporations in tea plantation business. Agri-business farms are mechanised, large in size, highly structured, reliant on chemicals, and may be described as 'agro-factories'.

Industries based on Inputs/Raw Materials

On the basis of the raw materials used, the industries are classified as: (a) agro-based; (b) mineral based; (c) chemical based; (d) forest based: and (e) animal based.

(a) *Agro based Industries*

Agro processing involves the processing of raw materials from the field and the farm into finished products for rural and urban markets. Major agro-processing industries are food processing, sugar, pickles, fruits juices, beverages (tea, coffee and cocoa), spices and oils fats and textiles (cotton, jute, silk), rubber, etc.

Food Processing

Agro processing includes canning, producing cream, fruit processing and confectionery. While some preserving techniques, such as drying, fermenting and pickling, have been known since ancient times, these had limited applications to cater to the pre-Industrial Revolution demands.

(b) *Mineral based Industries*

These industries use minerals as a raw material. Some industries use ferrous metallic minerals which contain ferrous (iron), such as iron and steel industries but some use non-ferrous metallic minerals, such as aluminium, copper and jewellery industries. Many industries use non-metallic minerals such as cement and pottery industries.

(c) *Chemical based Industries*

Such industries use natural chemical minerals, e.g. mineral-oil (petroleum) is used in petro-chemical industry. Salts, sulphur and potash industries also use natural minerals. Chemical industries are also based on raw materials obtained from wood and coal. Synthetic fibre, plastic, etc. are other examples of chemical based industries.

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(d) Forest based Raw Material using Industries

The forests provide many major and minor products which are used as raw material. Timber for furniture industry, wood, bamboo and grass for paper industry, lac for lac industries come from forests.

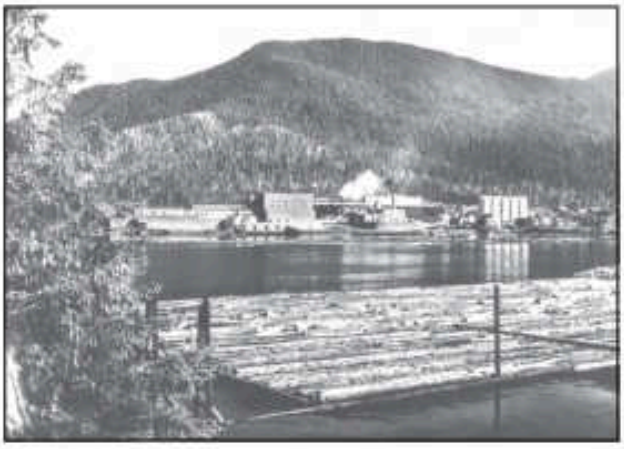


Fig. 5.6: A pulp mill in the heart of the Ketchikan's timber area of Alaska

(e) Animal based Industries

Leather for leather industry and wool for woollen textiles are obtained from animals. Besides, ivory is also obtained from elephant's tusks.

Industries Based On Output/Product

You have seen some machines and tools made of iron or steel. The raw material for such machines and tools is iron and steel. Which is itself an industry. The industry whose products are used to make other goods by using them as raw materials are basic industries. Can you identify the links? Iron/steel ⟶ machines for textile industry ⟶ clothes for use by consumers.

The consumer goods industries produced goods which are consumed by consumers directly. For example, industries producing breads and biscuits, tea, soaps and toiletries, paper for writing, televisions, etc. are consumer goods or non-basic industries.

INDUSTRIES BASED ON OWNERSHIP

(a) Public Sector Industries are owned and managed by governments. In India, there were a number of Public Sector Undertakings (PSUs). Socialist countries have many state owned industries. Mixed economies have both Public and Private sector enterprises.

(b) Private Sector Industries are owned by individual investors. These are managed by private organisations. In capitalist countries, industries are generally owned privately.

(c) Joint Sector Industries are managed by joint stock companies or sometimes the private and public sectors together establish and manage the industries. Can you make a list of such industries?

Concept of High Technology Industry

High technology, or simply high-tech, is the latest generation of manufacturing activities. It is best understood as the application of intensive research and development (R and D) efforts leading to the manufacture of products of an advanced scientific and engineering character. Professional (white collar) workers make up a large share of the total workforce. These highly skilled specialists greatly outnumber the actual production (blue collar) workers. Robotics on the assembly line, computer-aided design (CAD) and manufacturing, electronic controls of smelting and refining processes, and the constant development of new chemical and pharmaceutical products are notable examples of a high-tech industry.

Neatly spaced, low, modern, dispersed, office-plant-lab buildings rather than massive assembly structures, factories and storage areas mark the high-tech industrial landscape. Planned business parks for high-tech start-ups have become part of regional and local development schemes.

High-tech industries which are regionally concentrated, self-sustained and highly specialised are called technopolies.

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The Silicon Valley near San Francisco and Silicon Forest near Seattle are examples of technopolies. Are some technopolies developing in India?

Manufacturing contributes significantly to the world economy. Iron and steel, textiles, automobiles, petrochemicals and electronics are some of the world's most important manufacturing industries.



EXERCISES

1. Choose the right answer from the four alternatives given below.

(i) Which one of the following statements is wrong?

(a) Cheap water transport has facilitated the jute mill industry along the Hugli.

(b) Sugar, cotton textiles and vegetable oils are footloose industries.

(c) The development of hydro-electricity and petroleum reduced, to a great extent, the importance of coal energy as a locational factor for industry.

(d) Port towns in India have attracted industries.

(ii) In which one of the following types of economy are the factors of production owned individually ?

(a) Capitalist (c) Socialist

(b) Mixed (d) None

(iii) Which one of the following types of industries produces raw materials for other industries?

(a) Cottage Industries (c) Basic Industries

(b) Small-scale Industries (d) Footloose Industries

(iv) Which one of the following pairs is correctly matched ?

(a) Automobile industry ... Los Angeles

(b) Shipbuilding industry ... Lusaka

(c) Aircraft industry ... Florence

2. Write a short note on the following in about 30 words.

(i) High-Tech industry

(ii) Manufacturing

(iii) Footloose industries

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Unit-III

Chapter-6

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Tertiary and Quaternary Activities



When you fall ill you go to your family doctor or you call a doctor. Sometimes your parents take you to a hospital for treatment. While in school, you are taught by your teachers. In the event of any dispute, legal opinion is obtained from a lawyer. Likewise, there are many professionals who provide their services against payment of their fee. Thus, all types of services are special skills provided in exchange of payments. Health, education, law, governance and recreation etc. require professional skills. These services require other theoretical knowledge and practical training. Tertiary activities are related to the service sector. Manpower is an important component of the service sector as most of the tertiary activities are performed by skilled labour, professionally trained experts and consultants.

In the initial stages of economic development, larger proportion of people worked in the primary sector. In a developed economy, the majority of workers get employment in tertiary activity and a moderate proportion is employed in the secondary sector.

Tertiary activities include both production and exchange. The production involves the 'provision' of services that are 'consumed'. The output is indirectly measured in terms of wages and salaries. Exchange, involves trade, transport and communication facilities that are used to overcome distance. Tertiary activities, therefore, involve the commercial output of services rather than the production of tangible goods. They are not directly involved in the processing of physical raw materials. Common examples are the work of a plumber, electrician, technician, launderer, barber, shopkeeper, driver, cashier, teacher, doctor, lawyer and publisher etc. The main difference between secondary activities and tertiary activities is that the expertise provided by services relies more heavily on specialised skills, experience and knowledge of the workers rather than on the production techniques, machinery and factory processes.

TYPES OF TERTIARY ACTIVITIES

By now you know that you purchase your books, stationery from traders shop, travel by

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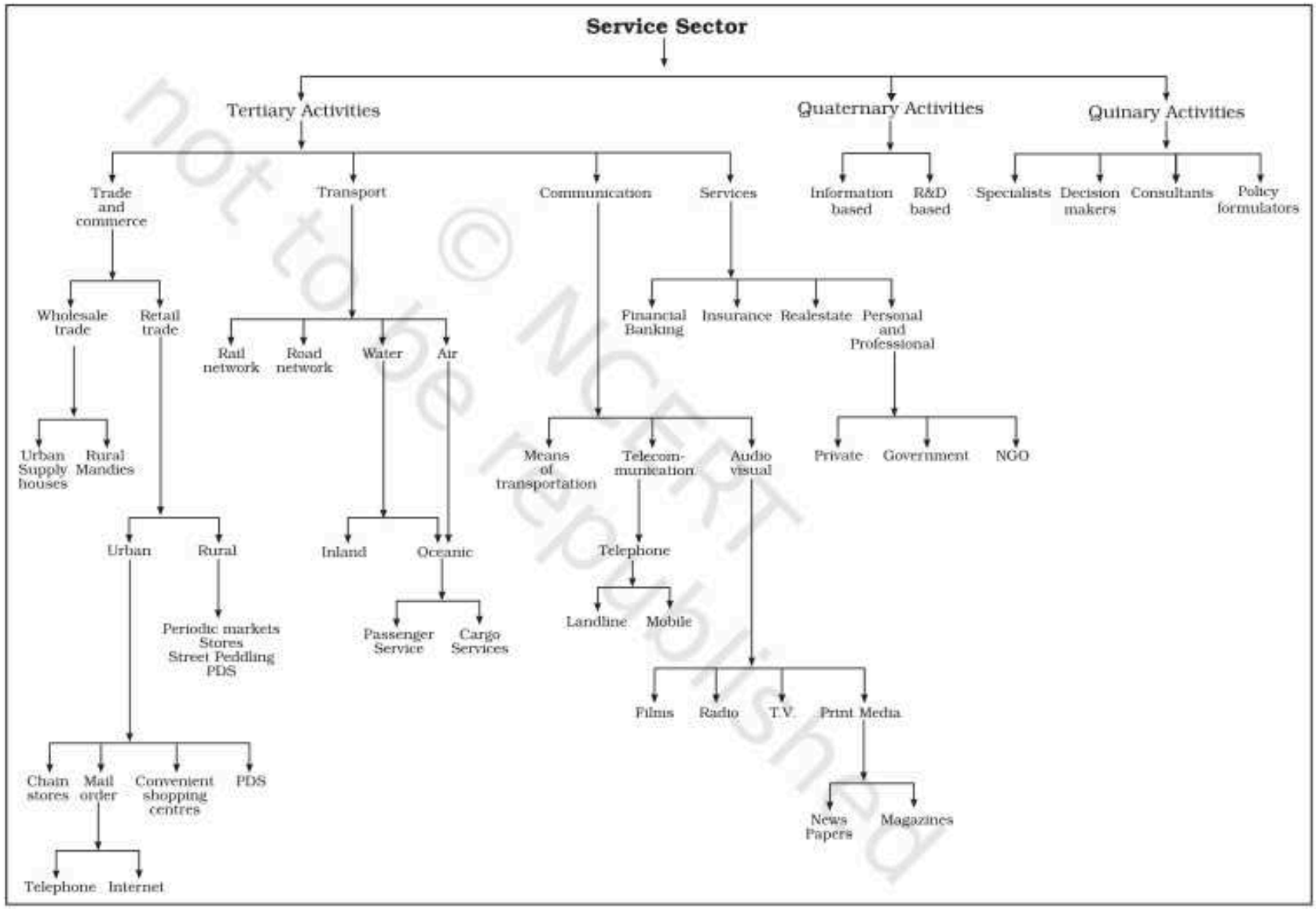


Fig. 6.1: Service Sector

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bus or rail, send letters, talk on telephone and obtain services of teachers for studies and doctors at the time of illness.

Thus, trade, transport, communication and services are some of the tertiary activities discussed in this section. The chart provides the basis for classifying the tertiary activities.

Trade and commerce

Trade is essentially **buying** and **selling** of items produced elsewhere. All the services in retail and wholesale trading or commerce are specifically intended for profit. The towns and cities where all these works take place are known us **trading centres.**

The rise of trading from barter at the local level to money-exchange of international scale has produced many centres and institutions such as **trading centres** or collection and distribution points.

Trading centres may be divided into rural and urban marketing centres.

Rural marketing centres cater to nearby settlements. These are quasi-urban centres. They serve as trading centres of the most rudimentary type. Here personal and professional services are not well-developed. These form local collecting and distributing centres. Most of these have *mandis* (wholesale markets) and also retailing areas. They are not urban centres *per se* but are significant centres for making available goods and services which are most frequently demanded by rural folk.



Fig. 6.2: A Wholesale Vegetable Market

Periodic markets in rural areas are found where there are no regular markets and local periodic markets are organised at different temporal intervals. These may be weekly, bi-weekly markets from where people from the surrounding areas meet their temporally accumulated demand. These markets are held on specified dates and move from one place to another. The shopkeepers thus, remain busy on all the days while a large area is served by them.

Urban marketing centres have more widely specialised urban services. They provide ordinary goods and services as well as many of the specialised goods and services required by people. Urban centres, therefore, offer manufactured goods as well as many specialised markets develop, e.g. markets for labour, housing, semi or finished products. Services of educational institutions and professionals such as teachers, lawyers, consultants, physicians, dentists and veterinary doctors are available.



Fig. 6.3: Packed Food Market in U.S.A.

Retail Trading

This is the business activity concerned with the sale of goods directly to the consumers. Most of the retail trading takes place in fixed establishments or stores solely devoted to selling. Street peddling, handcarts, trucks, door-to-door, mail-order, telephone, automatic vending machines and internet are examples of non-store retail trading.

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More on Stores

Consumer cooperatives were the first of the large-scale innovations in retailing.

Departmental stores delegate the responsibility and authority to departmental heads for purchasing of commodities and for overseeing the sale in different sections of the stores.

Chain stores are able to purchase merchandise most economically, often going so far as to direct the goods to be manufactured to their specification. They employ highly skilled specialists in many executive tasks. They have the ability to experiment in one store and apply the results to many.

Wholesale Trading

Wholesale trading constitutes bulk business through numerous intermediary merchants and supply houses and not through retail stores. Some large stores including chain stores are able to buy directly from the manufacturers. However, most retail stores procure supplies from an intermediary source. Wholesalers often extend credit to retail stores to such an extent that the retailer operates very largely on the wholesaler's capital.

Transport

Transport is a service or facility by which people, materials and manufactured goods are physically carried from one location to another. It is an organised industry created to satisfy man's basic need of mobility. Modern society requires speedy and efficient transport systems to assist in the production, distribution and consumption of goods. At every stage in this complex system, the value of the material is significantly enhanced by transportation.

Transport distance can be measured as: **km distance** or actual distance of route length; **time distance** or the time taken to travel on a particular route; and **cost distance** or the expense of travelling on a route. In selecting the mode of transport, distance, in terms of time or cost, is the determining factor. Isochrone lines are drawn on a map to join places equal in terms of the time taken to reach them.

Network and Accessibility

As transport systems develop, different places are linked together to form a **network.** Networks are made up of nodes and links. A **node** is the meeting point of two or more routes, a point of origin, a point of destination or any sizeable town along a route, Every road that joins two nodes is called a **link.** A developed network has many links, which means that places are well-connected.

Factors Affecting Transport

Demand for transport is influenced by the size of population. The larger the population size, the greater is the demand for transport.

Routes depend on: location of cities, towns, villages, industrial centres and raw materials, pattern of trade between them, nature of the landscape between them, type of climate, and funds available for overcoming obstacles along the length of the route.

Communication

Communication services involve the transmission of **words** and **messages, facts** and **ideas.** The invention of writing preserved messages and helped to make communication dependent on means of transport. These were actually carried by hand, animals, boat, road, rail and air. That is why all forms of transport are also referred to as lines of communication. Where the transport network is efficient, communications are easily disseminated. Certain developments, such as mobile telephony and satellites, have made communications independent of transport. All forms are not fully disassociated because of the cheapness of the older systems. Thus, very

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large volumes of mail continue to be handled by post offices all over the world.

Some of the communication services are discussed below.

Telecommunications

The use of telecommunications is linked to the development of modern technology. It has revolutionised communications because of the speed with which messages are sent. The time reduced is from weeks to minutes. Besides, the recent advancements like mobile telephony have made communications direct and instantaneous at any time and from anywhere. The telegraph, morse code and telex have almost become things of the past.

Radio and **television** also help to relay news, pictures, and telephone calls to vast audiences around the world and hence they are termed as **mass media.** They are vital for advertising and entertainment. **Newspapers** are able to cover events in all corners of the world. Satellite communication relays information of the earth and from space. The **internet** has truly revolutionised the global communication system .

Services

Services occur at many different levels. Some are geared to industry, some to people, and some to both industry and people, e.g. the transport systems. Low-order services, such as grocery shops and laundries, are more common and widespread than high-order services or more specialised ones like those of accountants, consultants and physicians. Services are provided to individual consumers who can afford to pay for them. For example, the gardener, the launderers and the barber do primarily physical labour. Teacher, lawyers, physicians, musicians and others perform mental labour.

Many services have now been regulated. Making and maintaining highways and bridges, maintaining fire fighting departments and supplying or supervising education and customer-care are among the important services most often supervised or performed by governments or companies. State and union legislation have established corporations to supervise and control the marketing of such services as transport, telecommunication, energy and water supply. Professional services are primarily health care, engineering, law and management. The location of recreational and entertainment services depends on the market. Multiplexes and restaurants might find location within or near the Central Business District (CBD), whereas a golf course would choose a site where land costs are lower than in the CBD.

Personal services are made available to the people to facilitate their work in daily life. The workers migrate from rural areas in search of employment and are unskilled. They are employed in domestic services as housekeepers, cooks, and gardeners. This segment of workers is generally unorganised. One such example in India is Mumbai's *dabbawala* (Tiffin) service provided to about 1,75,000 customers all over the city.



Fig. 6.4: Dabbawala Service in Mumbai

PEOPLE ENGAGED IN TERTIARY ACTIVITIES

Today most people are service workers. Services are provided in all societies. But in more developed countries a higher percentage of workers is employed in providing services as compared to less developed countries. The trend in employment in this sector has been

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increasing while it has remained unchanged or decreasing in the primary and secondary activities.

SOME SELECTED EXAMPLES

Tourism

Tourism is travel undertaken for purposes of recreation rather than business. It has become the world's single largest tertiary activity in total registered jobs (250 million) and total revenue (40 per cent of the total GDP). Besides, many local persons, are employed to provide services like accommodation, meals, transport, entertainment and special shops serving the tourists. Tourism fosters the growth of infrastructure industries, retail trading, and craft industries (souvenirs). In some regions, tourism is seasonal because the vacation period is dependent on favourable weather conditions, but many regions attract visitors all the year round.



Fig. 6.5: Tourists skiing in the snow capped mountain slopes of Switzerland

Tourist Regions

The warmer places around the Mediterranean Coast and the West Coast of India are some of the popular tourist destinations in the world. Others include winter sports regions, found mainly in mountainous areas, and various scenic landscapes and national parks, which are scattered. Historic towns also attract tourists, because of the monument, heritage sites and cultural activities.

Tourist Attractions

Climate: Most people from colder regions expect to have warm, sunny weather for beach holidays. This is one of the main reasons for the importance of tourism in Southern Europe and the Mediterranean lands. The Mediterranean climate offers almost consistently higher temperatures, than in other parts of Europe, long hours of sunshine and low rainfall throughout the peak holiday season. People taking winter holidays have specific climatic requirements, either higher temperatures than their own homelands, or snow cover suitable for skiing.

Landscape: Many people like to spend their holidays in an attractive environment, which often means mountains, lakes, spectacular sea coasts and landscapes not completely altered by man.

History and Art: The history and art of an area have potential attractiveness. People visit ancient or picturesque towns and archaeological sites, and enjoy exploring castles, palaces and churches.

Culture and Economy: These attract tourists with a penchant for experiencing ethnic and local customs. Besides, if a region provides for the needs of tourists at a cheap cost, it is likely to become very popular. Home-stay has emerged as a profitable business such as *heritage homes* in Goa, Madikere and Coorg in Karnataka.

Medical Services for Overseas Patients in India

About 55,000 patients from U.S.A. visited India in 2005 for treatment. This is still a small number compared with the millions of surgeries performed each year in the U.S. healthcare system. India has emerged as the leading country of medical tourism in the world. World class hospitals located in metropolitan cities cater to patients all over the world. Medical tourism brings abundant benefits to developing countries like India, Thailand, Singapore and Malaysia. Beyond medical tourism, is the trend

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of outsourcing of medical tests and data interpretation. Hospitals in India, Switzerland and Australia have been performing certain medical services – ranging from reading radiology images, to interpreting Magnetic Resonance Images (MRIs) and ultrasound tests. Outsourcing holds tremendous advantages for patients, if it is focused on improving quality or providing specialised care.

Medical Tourism

When medical treatment is combined with international tourism activity, it lends itself to what is commonly known as medical tourism.

QUATERNARY ACTIVITIES

What do a CEO of an MNC in Copenhagen, at New York and a medical transcriptionist at Bangalore have in common? All these people work in a segment of the service sector that is knowledge oriented. This sector can be divided into quaternary and quinary activities.

Quaternary activities involve some of the following: the collection, production and dissemination of information or even the production of information. Quaternary activities centre around research, development and may be seen as an advanced form of services involving specialised knowledge and technical skills.

The Quaternary Sector

The Quaternary Sector along with the Tertiary Sector has replaced most of the primary and secondary employment as the basis for economic growth. Over half of all workers in developed economies are in the 'Knowledge Sector' and there has been a very high growth in demand for and consumption of information-based services from mutual fund managers to tax consultants, software developers and statisticians. Personnel working in office buildings, elementary schools and university classrooms, hospitals and doctors' offices, theatres, accounting and brokerage firms all belong to this category of services.

Like some of the tertiary functions, quaternary activities can also be outsourced. They are not tied to resources, affected by the environment, or necessarily localised by market.

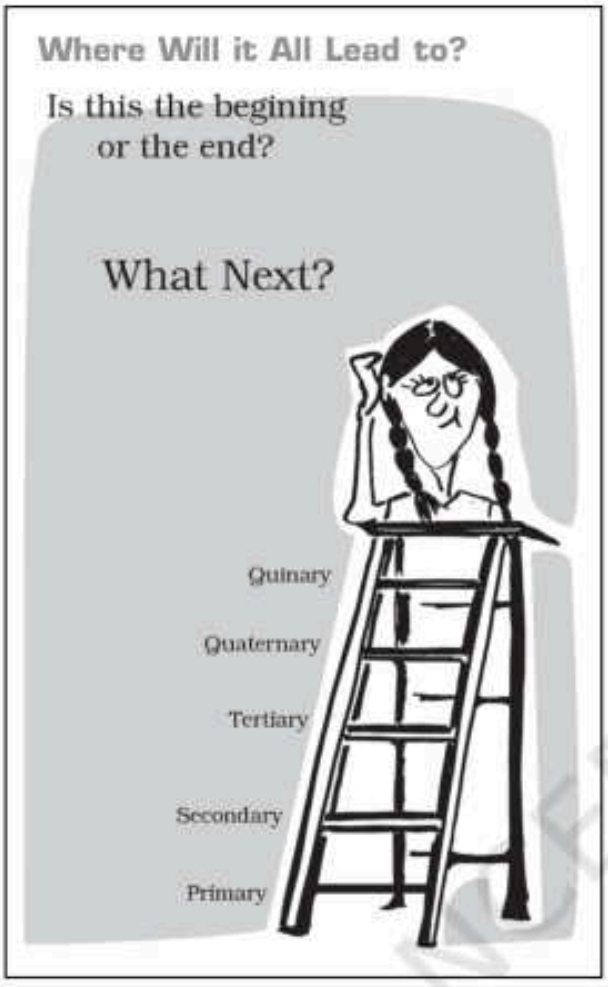
QUINARY ACTIVITIES

The highest level of decision makers or policy makers perform quinary activities. These are subtly different from the knowledge based industries that the quinary sector in general deals with.

Quinary activities are services that focus on the creation, re-arrangement and interpretation of new and existing ideas; data interpretation and the use and evaluation of new technologies. Often referred to as 'gold collar' professions, they represent another subdivision of the tertiary sector representing special and highly paid skills of senior business executives, government officials, research scientists, financial and legal consultants, etc. Their importance in the structure of advanced economies far outweighs their numbers.

Outsourcing has resulted in the opening up of a large number of call centres in India, China, Eastern Europe, Israel, Philippines and Costa Rica. It has created new jobs in these countries. Outsourcing is coming to those countries where cheap and skilled workers are available. These are also out-migrating countries. With the work available though outsourcing, the migration in these countries may come down. Outsourcing countries are facing resistance from job-seeking youths in their respective countries. The comparative advantage is the main reason for continuing outsourcing. New trends in quinary services include knowledge processing outsourcing (KPO) and 'home shoring', the latter as an alternative to outsourcing. The KPO industry is distinct from Business Process Outsourcing (BPO) as it involves highly skilled workers. It is information driven knowledge outsourcing. KPO enables companies to create additional business opportunities. Examples of KPOs include research and development (R and D) activities, e-learning, business research, intellectual property (IP) research, legal profession and the banking sector.

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Outsourcing

Outsourcing or contracting out is giving work to an outside agency to improve efficiency and to reduce costs. When outsourcing involves transferring work to overseas locations, it is described by the term off - shoring, although both off - shoring and outsourcing are used together. Business activities that are outsourced include information technology (IT), human resources, customer support and call centre services and at times also manufacturing and engineering.

Data processing is an IT related service easily be carried out in Asian, East European and African countries, In these countries IT skilled staff with good English language skills are available at lower wages than those in the developed countries. Thus, a company in Hyderabad or Manila does work on a project based on GIS techniques for a country like U.S.A or Japan. Overhead costs are also much lower making it profitable to get job-work carried out overseas, whether it is in India, China or even a less populous country like Botswana in Africa.

Activity

Describe the nature of work against each colour-name

Colour of the collar	*Nature of work*
Red	?
Gold	?
White	?
Grey	?
Blue	?
Pink	?

THE DIGITAL DIVIDE

Opportunities emerging from the Information and Communication Technology based development is unevenly distributed across the globe. There are wide ranging economic, political and social differences among countries. How quickly countries can provide ICT access and benefits to its citizens is the deciding factor. While developed countries in general have surged forward, the developing countries have lagged behind and this is known as the digital divide. Similarly digital divides exist within countries. For example, in a large country like India or Russia, it is inevitable that certain areas like metropolitan centres possess better connectivity and access to the digital world versus peripheral rural areas.

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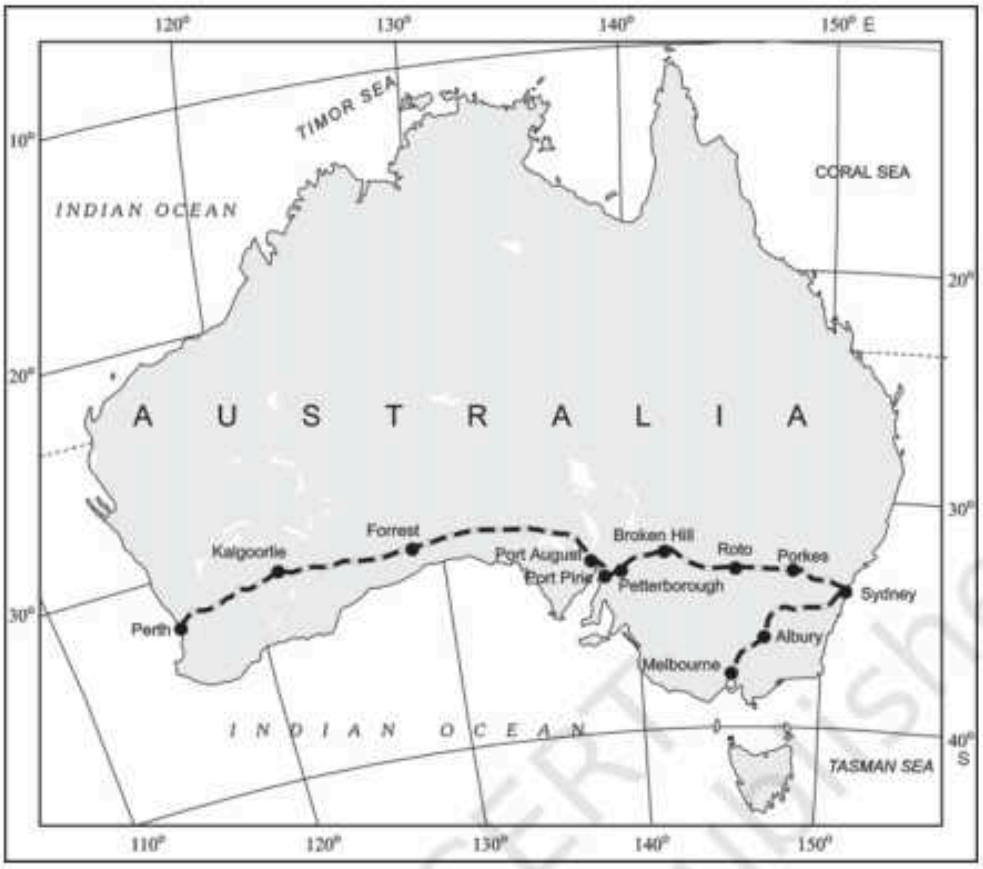


Fig. 7.7: Australian Trans-Continental Railway



Fig. 7.8: The view of Seine River from the Eiffel Tower (One can see how the river has become an important Inland waterway)

Sea Routes

The oceans offer a smooth highway traversable in all directions with no maintenance costs. Its transformation into a routeway by sea-going vessels is an important development in human adaptation to the physical environment. Compared to land and air, ocean transport is a cheaper means of haulage (carrying of load) of bulky material over long distances from one continent to another.

Modern passenger liners (ships) and cargo ships are equipped with radar, wireless and other navigation aids. The development of refrigerated chambers for perishable goods, tankers and specialised ships has also improved cargo transport. The use of containers has made cargo handling at the world's major ports easier.

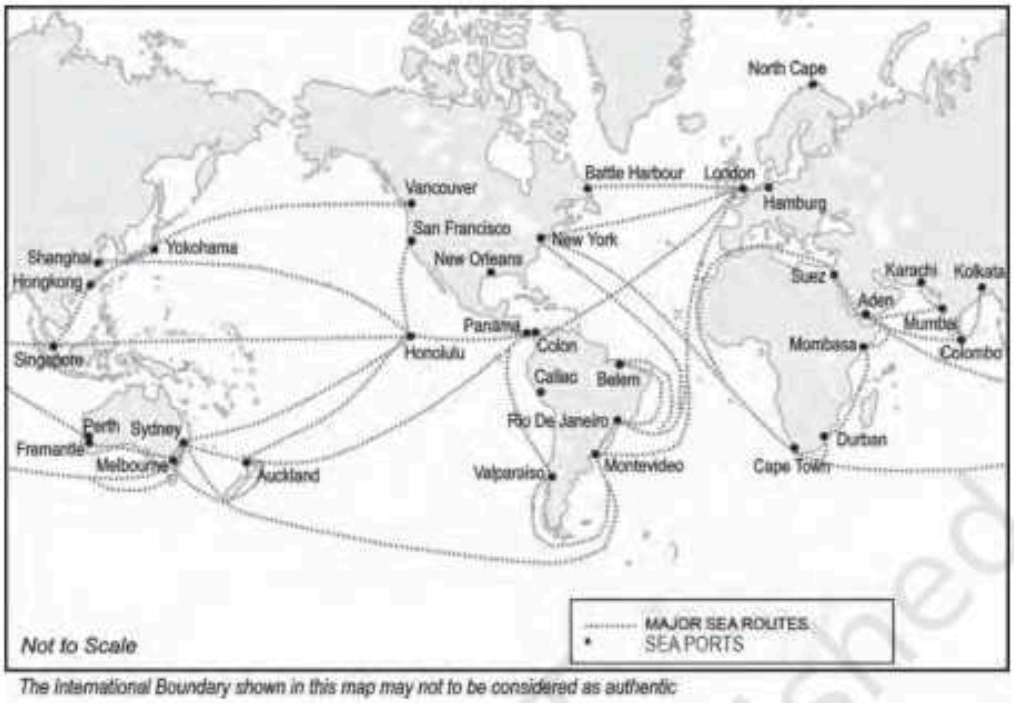
Important Sea Routes

Major sea routes are shown in the Fig. 7.9. Some important routes have been discussed in the following pages.

The Northern Atlantic Sea Route

This links North-eastern U.S.A. and North-western Europe, the two industrially developed

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The International Boundary shown in this map may not to be considered as authentic

Fig. 7.9: Major Sea Routes and Sea Ports

regions of the world. The foreign trade over this route is greater than that of the rest of the world combined. One fourth of the world's foreign trade moves on this route. It is, therefore, the busiest in the world and otherwise, called the Big Trunk Route. Both the coasts have highly advanced ports and harbour facilities.

Activity

Find out some of the important ports on the coast of U.S.A. and Western Europe in your atlas.

The Mediterranean–Indian Ocean Sea Route

This sea route passes through the heart of the Old World and serves more countries and people than any other route. Port Said, Aden, Mumbai, Colombo and Singapore are some of the important ports on this route. The construction of Suez Canal has greatly reduced the distance and time as compared to the earlier route through the Cape of Good Hope, which was longer than the route through Suez Canal.

The Cape of Good Hope Sea Route

This trade route connects the highly industrialised Western European region with West Africa, South Africa, South-east Asia and the commercial agriculture and livestock economies of Australia and New Zealand. The volume of trade and traffic between both East and West Africa is on the increase due to the development of the rich natural resources such as gold, diamond, copper, tin, groundnut, oil palm, coffee and fruits.

The Southern Atlantic Sea Route

This sea route is another important one across the Atlantic Ocean which connects West European and West African countries with Brazil, Argentina and Uruguay in South America. The traffic is far less on this route because of the limited development and population in South America and Africa. Only southeastern Brazil and Plata estuary and parts of South Africa have large-scale industries. There is also little traffic on the route between Rio de Janeiro and Cape Town because both South America and Africa have similar products and resources.

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The North Pacific Sea Route

Trade across the vast North Pacific Ocean moves by several routes which converge at Honolulu. The direct route on the Great Circle links Vancouver and Yokohama and reduces the travelling distance (2,480 km) by half.

This sea route links the ports on the west-coast of North America with those of Asia. These are Vancouver, Seattle, Portland, San Francisco and Los Angeles on the American side and Yokohama, Kobe, Shanghai, Hong Kong, Manila and Singapore on the Asian side.

The South Pacific Sea Route

This sea route connects Western Europe and North America with Australia, New Zealand and the scattered Pacific islands via the Panama Canal. This route is also used for reaching Hong Kong, Philippines and Indonesia. The distance covered between Panama and Sydney is 12,000 km. Honolulu is an important port on this route.

Coastal Shipping

It is obvious that water transport is a cheaper mode. While oceanic routes connect different countries, coastal shipping is a convenient mode of transportation with long coastlines, e.g. U.S.A, China and India. Shenzhen States in Europe are most suitably placed for coastal shipping connecting one member's coast with the other. If properly developed, coastal shipping can reduce the congestion on the land routes.

Shipping Canals

The Suez and the Panama Canals are two vital man-made navigation canals or waterways which serve as gateways of commerce for both the eastern and western worlds.

The Suez Canal

This canal had been constructed in 1869 in Egypt between Port Said in the north and Port Suez in the south linking the Mediterranean Sea and the Red Sea. It gives Europe a new gateway to the Indian Ocean and reduces direct sea-route distance between Liverpool and Colombo compared to the Cape of Good Hope route. It is a sea-level canal without locks which is about 160 km and 11 to 15 m deep. About 100 ships travel daily and each ship takes 10-12 hours to cross this canal. The tolls are so heavy that some find it cheaper to go by the longer Cape Route whenever the consequent delay is not important. A railway follows the canal to Suez, and from Ismailia there is a branch line to Cairo. A navigable fresh-water canal from the Nile also joins the Suez Canal in Ismailia to supply fresh-water to Port Said and Suez.

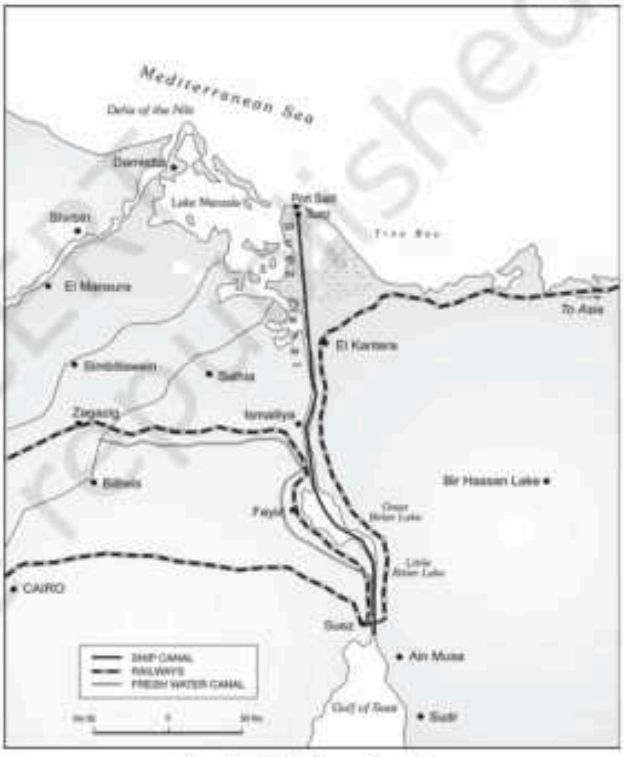


Fig. 7.10 : Suez Canal

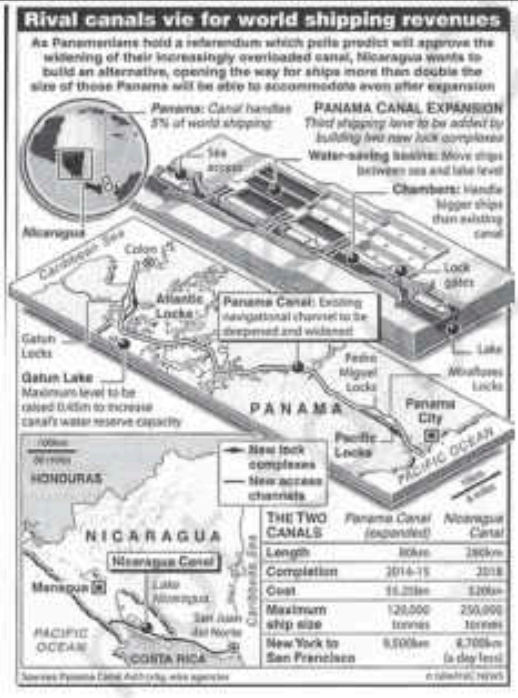
The Panama Canal

This canal connects the Atlantic Ocean in the east to the Pacific Ocean in the west. It has been constructed across the Panama Isthmus between Panama City and Colon by the U.S. government which purchased 8 km of area on either side and named it the Canal Zone. The Canal is about 72 km. long and involves a very

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Fig. 7.11 : The Panama Canal



Activity

Can you think of the impact on traffic in Panama canal after the Nicaraguan canal opens up?

deep cutting for a length of 12 km. It has a six-lock system and ships cross the different levels (26 m up and down) through these locks before entering the Gulf of Panama.

It shortens the distance between New York and San Francisco by 13,000 km by sea. Likewise the distance between Western Europe and the West-coast of U.S.A.; and North-eastern and Central U.S.A. and East and South-east Asia is shortened. The economic significance of this Canal is relatively less than that of the Suez. However, it is vital to the economies of Latin America.

Inland Waterways

Rivers, canals, lakes and coastal areas have been important waterways since time immemorial. Boats and steamers are used as means of transport for cargo and passengers. The development of inland waterways is dependent on the **navigability** width and depth of the channel, continuity in the **water flow**, and **transport technology** in use. Rivers are the only means of transport in dense forests. Very heavy cargo like coal, cement, timber and metallic ores can be transported through inland waterways. In ancient times, riverways were the main highways of transportation as in the case of India. But they lost importance because of competition from railways, lack of water due to diversion for irrigation, and their poor maintenance.



Fig. 7.12: Inland waterways are a major source of transport wherever the river is wide, deep and free of silt

The significance of rivers as inland waterways for domestic and international

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States/UTs of large size and large population

States/UTs of large size but small population

States/UTs of smaller size but larger population

Check from the table (Appendix–iA) that U.P., Maharashtra, Bihar, West Bengal, Andhra Pradesh along with Tamil Nadu, Madhya Pradesh, Rajasthan, Karnataka and Gujarat, together account for about 76 per cent of the total population of the country. On the other hand, share of population is very small in the states like Jammu & Kashmir (1.04%), Arunachal Pradesh (0.11%) and Uttarakhand (0.84%) inspite of theses states having fairly large geographical area.

Such an uneven spatial distribution of population in India suggests a close relationship between population and physical, socio-economic and historical factors. As far as the physical factors are concerned, it is clear that climate along with terrain and availability of water largely determines the pattern of the population distribution. Consequently, we observe that the North Indian Plains, deltas and Coastal Plains have higher proportion of population than the interior districts of southern and central Indian States, Himalayas, some of the north eastern and the western states. However, development of irrigation (Rajasthan), availability of mineral and energy resources (Jharkhand) and development of transport network (Peninsular States) have resulted in moderate to high concentration of population in areas which were previously very thinly populated.

Among the socio-economic and historical factors of distribution of population, important ones are evolution of settled agriculture and agricultural development; pattern of human settlement; development of transport network, industrialisation and urbanisation. It is observed that the regions falling in the river plains and coastal areas of India have remained the regions of larger population concentration. Even though the uses of natural resources like land and water in these regions have shown the sign of degradation, the concentration of population remains high because of an early history of human settlement and development of transport network. On the other hand, the urban regions of Delhi, Mumbai, Kolkata, Bengaluru, Pune, Ahmedabad, Chennai and Jaipur have high concentration of population due to industrial development and urbanisation drawing a large numbers of rural-urban migrants.

Density of Population

Density of population, is expressed as number of persons per unit area. It helps in getting a better understanding of the spatial distribution of population in relation to land. The density of population in India (2011) is 382 persons per sq km. There has been a steady increase of more than 200 persons per sq km over the last 50 years as the density of population increased from 117 persons/ sq km in 1951 to 382 persons/sq km in 2011.

The data shown in Appendix (i) give an idea of spatial variation of population densities in the country which ranges from as low as 17 persons per sq km in Arunachal Pradesh to 11,297 persons in the National Capital Territory of Delhi. Among the northern Indian States, Bihar (1102), West Bengal (1029) and and Uttar Pradesh (828) have higher densities, while Kerala (859) and Tamil Nadu (555) have higher densities among the peninsular Indian states. States like Assam, Gujarat, Andhra Pradesh, Haryana, Jharkhand, Odisha have moderate densities. The hill states of the Himalayan region and North eastern states of India (excluding Assam) have relatively low densities while the Union Territories (excluding Andaman and Nicobar islands) have very high densities of population (Appendix–i).

The density of population, as discussed in the earlier paragraph, is a crude measure of human and land relationship. To get a better insight into the human-land ratio in terms of pressure of population on total cultivable land, the *physiological* and the *agricultural* densities should be found out which are significant for a country like India having a large agricultural population.



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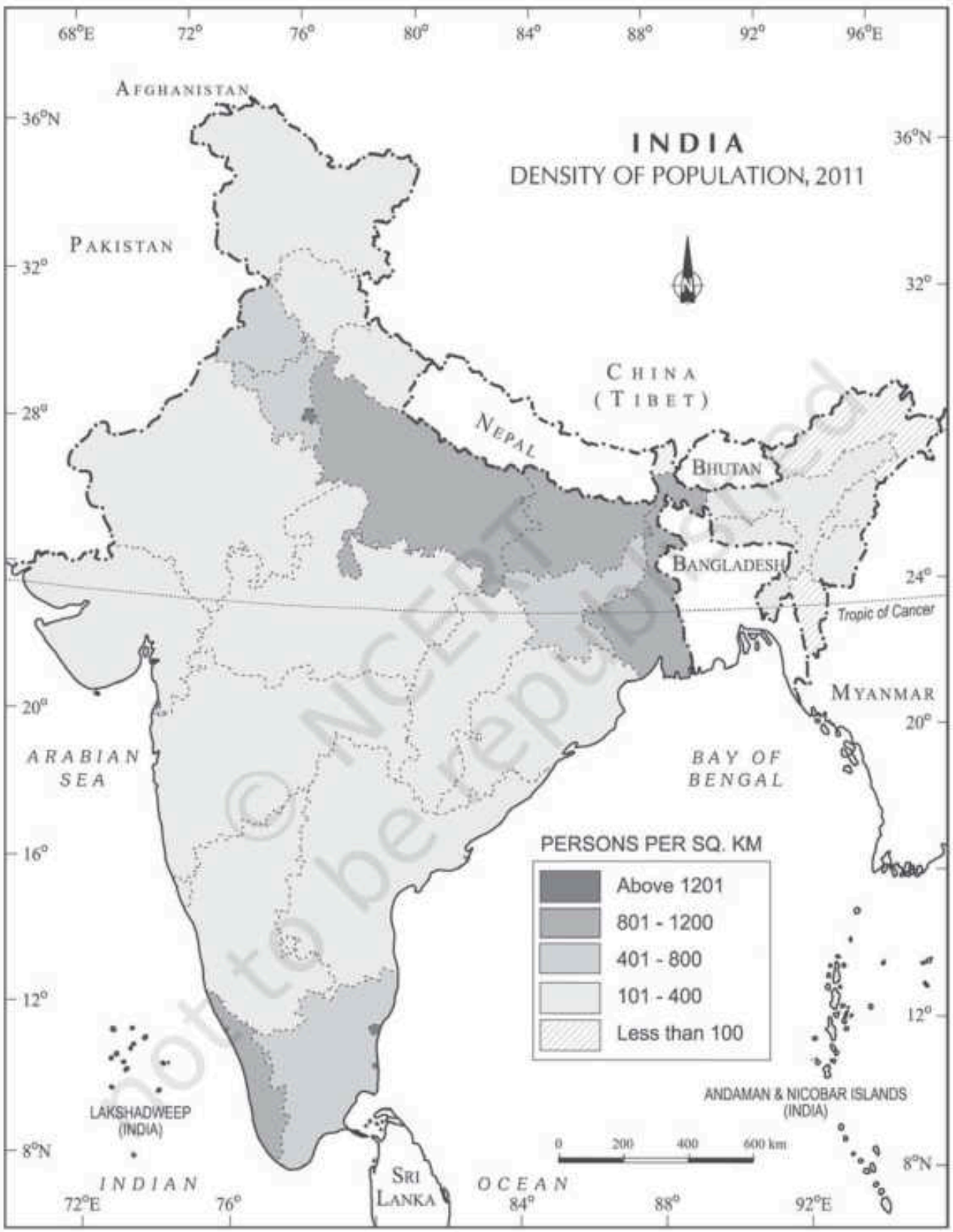


Fig. 1.2 : India – Density of Population

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Physiological density = total population / net cultivated area

Agricultural density = total agricultural population / net cultivable area

Agricultural population includes cultivators and agricultural labourers and their family members.

Growth of Population

Growth of population is the change in the number of people living in a particular area between two points of time. Its rate is expressed in percentage. Population growth has two components namely; natural and induced. While the natural growth is analysed by assessing the crude birth and death rates, the induced components are explained by the volume of inward and outward movement of people in any given area. However, in the present chapter, we will only discuss the natural growth of India's population.

The decadal and annual growth rates of population in India are both very high and steadily increasing over time. The annual growth rate of India's population is 1.64 per cent (2011).

Population Doubling Time

Population doubling time is the time taken by any population to double itself at its current annual growth rate.

The growth rate of population in India over the last one century has been caused by annual birth rate and death rate and rate of migration and thereby shows different trends. There are four distinct phases of growth identified within this period:

Table 1.1 : Decadal Growth Rates in India, 1901-2011

Census Years	Total Population	Growth Rate*	
		Absolute Number	% of Growth
1901	238396327	----------	----------
1911	252093390	(+) 13697063	(+) 5.75
1921	251321213	(–) 772117	(–) 0.31
1931	278977238	(+) 27656025	(+) 11.60
1941	318660580	(+) 39683342	(+) 14.22
1951	361088090	(+) 42420485	(+) 13.31
1961	439234771	(+) 77682873	(+) 21.51
1971	548159652	(+) 108924881	(+) 24.80
1981	683329097	(+) 135169445	(+) 24.66
1991	846302688	(+) 162973591	(+) 23.85
2001	1028610328	(+) 182307640	(+) 21.54
2011**	1210193422	(+) 181583094	(+) 17.64

* Decadal growth rate: $g = \frac{P_2 - P_1}{P_1} 100$

where p_1 = population of the base year
p_2 = population of the present year

** Source : Census of India, 2011(Provisional)

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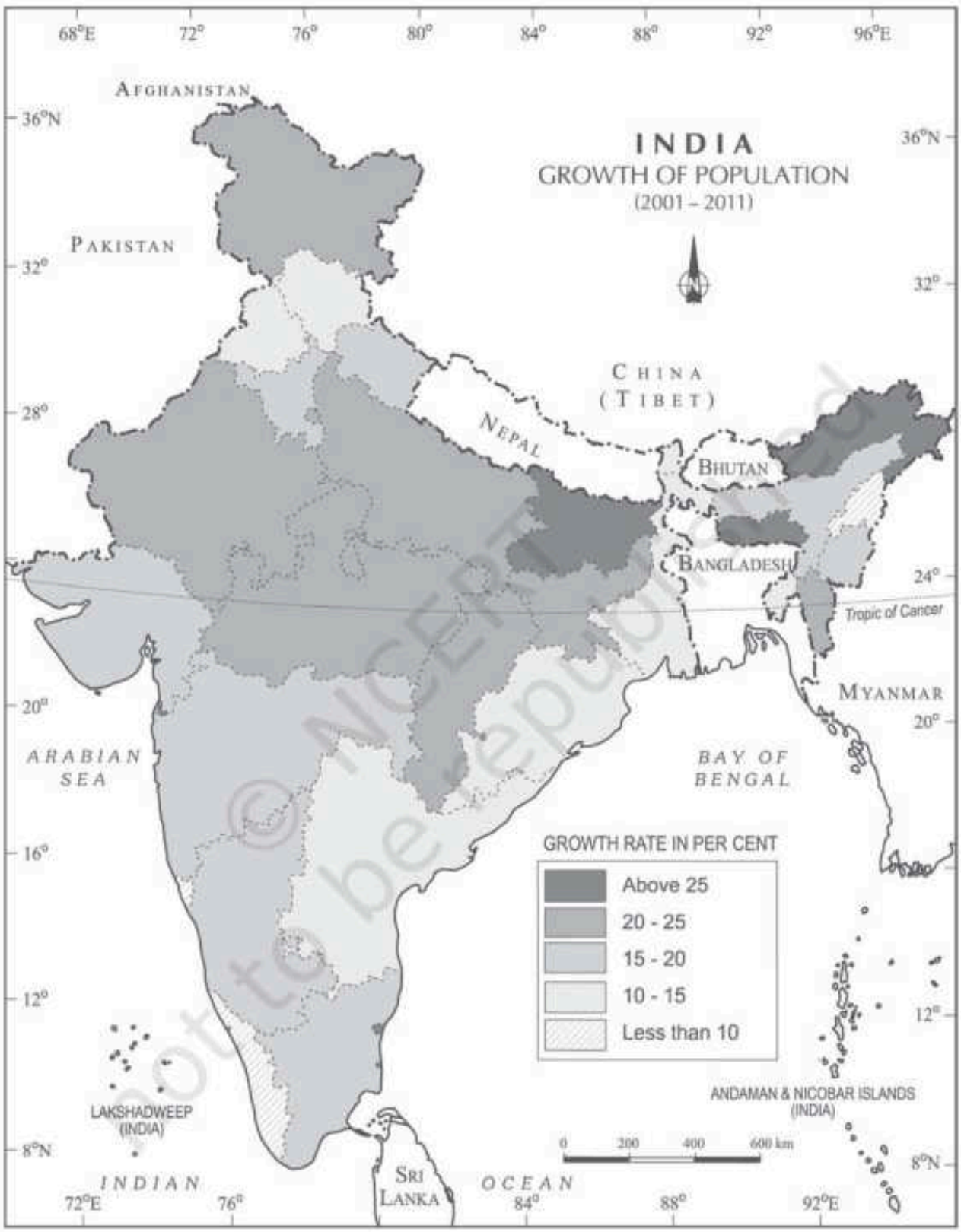


Fig. 1.3 : India – Growth of Population

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Phase I : The period from 1901-1921 is referred to as a period of stagnant or stationary phase of growth of India's population, since in this period growth rate was very low, even recording a negative growth rate during 1911-1921. Both the birth rate and death rate were high keeping the rate of increase low (Appendix–iii). Poor health and medical services, illiteracy of people at large and inefficient distribution system of food and other basic necessities were largely responsible for a high birth and death rates in this period.

Phase II : The decades 1921-1951 are referred to as the period of steady population growth. An overall improvement in health and sanitation throughout the country brought down the mortality rate. At the same time better transport and communication system improved distribution system. The crude birth rate remained high in this period leading to higher growth rate than the previous phase. This is impressive at the backdrop of Great Economic Depression, 1920s and World War II.

Phase III : The decades 1951-1981 are referred to as the period of population explosion in India, which was caused by a rapid fall in the mortality rate but a high fertility rate of population in the country. The average annual growth rate was as high as 2.2 per cent. It is in this period, after the Independence, that developmental activities were introduced through a centralised planning process and economy started showing up ensuring the improvement of living condition of people at large. Consequently, there was a high natural increase and higher growth rate. Besides, increased international migration bringing in Tibetans, Bangladeshis, Nepalies and even people from Pakistan contributed to the high growth rate.

Phase IV : In the post 1981 till present, the growth rate of country's population though remained high, has started slowing down gradually (Table 1.1). A downward trend of crude birth rate is held responsible for such a population growth. This was, in turn, affected by an increase in the mean age at marriage, improved quality of life particularly education of females in the country.

The growth rate of population is, however, still high in the country, and it has been projected by World Development Report that population of India will touch 1,350 million by 2025.

The analysis done so far shows the average growth rate, but the country also has wide variation (Appendix–iv) in growth rates from one area to another which is discussed below.

Regional Variation in Population Growth

The growth rate of population during 1991-2001 in Indian States and Union Territories shows very obvious pattern.

The States like Kerala, Karnataka, Tamil Nadu, Andhra Pradesh, Odisha, Puducherry, and Goa show a low rate of growth not exceeding 20 per cent over the decade. Kerala registered the lowest growth rate (9.4) not only in this group of states but also in the country as a whole.

A continuous belt of states from west to east in the north-west, north, and north central parts of the country has relatively high growth rate than the southern states. It is in this belt comprising Gujarat, Maharashtra, Rajasthan, Punjab, Haryana, Uttar Pradesh, Uttarakhand, Madhya Pradesh, Sikkim, Assam, West Bengal, Bihar, Chhattisgarh, and Jharkhand, the growth rate on the average remained 20-25 per cent.

During 2001-2011, the growth rates of almost all States and Union Territories have registered a lower figure compared to the previous decade, namely, 1991-2001. The percentage decadal growth rates of the six most populous States, namely, Uttar Pradesh, Maharashtra, Bihar, West Bengal, Andhra

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Pradesh and Madhya Pradesh have all fallen during 2001-2011 compared to 1991-2001, the fall being the lowest for Andhra Pradesh (3.5% percentage points) and highest for Maharashtra (6.7 percentage points). Tamil Nadu (3.9 percentage points) and Puducherry (7.1 percentage points) have registered some increase during 2001-2011 over the previous decade.

Activity

Take the population growth data of the districts/selected districts of your respective state for total male and female population and represent them with the help of Composite Bar Graph.

An important aspect of population growth in India is the growth of its adolescents. At present the share of adolescents i.e., up to the age group of 10-19 years is about 20.9 per cent (2011), among which male adolescents constitute 52.7 per cent and female adolescents constitute 47.3 per cent. The adolescent population, though, regarded as the youthful population having high potentials, but at the same time they are quite vulnerable if not guided and channelised properly. There are many challenges for the society as far as these adolescents are concerned, some of which are lower age at marriage, illiteracy – particularly female illiteracy, school dropouts, low intake of nutrients, high rate of maternal mortality of adolescent mothers, high rate of HIV and AIDS infections, physical and mental disability or retardedness, drug abuse and alcoholism, juvenile delinquency and commitence of crimes, etc.

In view of these, the Government of India has undertaken certain policies to impart proper education to the adolescent groups so that their talents are better channelised and properly utilised. The National Youth Policy is one example which has been designed to look into the overall development of our large youth and adolescent population.

The National Youth Policy (NYP–2014) launched in February 2014 proposes a holistic 'vision' for the youth of India, which is "To empower the youth of the country to achieve their full potential, and through them enable India to find its rightful place in the community of nations". The NYP–2014 has defined 'youth'as persons in the age group of 15–29 years.

The Government of India also formulated the National Policy for Skill Development and Entrepreneurship in 2015 to provide an umbrella framework to all skilling activities being carried out within the country, and to align these to common standards and link skilling with demand centres.

It appears from the above discussion that the growth rate of population is widely variant over space and time in the country and also highlights various social problems related to the growth of population. However, in order to have a better insight into the growth pattern of population it is also necessary to look into the social composition of population.

Population Composition

Population composition is a distinct field of study within population geography with a vast coverage of analysis of age and sex, place of residence, ethnic characteristics, tribes, language, religion, marital status, literacy and education, occupational characteristics, etc. In this section, the composition of Indian population with respect to their rural-urban characteristics, language, religion and pattern of occupation will be discussed.

Rural – Urban Composition

Composition of population by their respective places of residence is an important indicator of social and economic characteristics. This becomes even more significant for a country where about 68.8 per cent of its total population lives in village (2011).

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Do you know that India has 640,867 villages according to the Census 2011 out of which 597,608 (93.2 per cent) are inhabited villages? However, the distribution of rural population is not uniform throughout the country. You might have noted that the states like Himachal Pradesh and Bihar have very high percentage of rural population. The states of Goa and Mizoram have only little over half of their total population residing in villages.

The Union Territories, on the other hand, have smaller proportion of rural population, except Dadra and Nagar Haveli (53.38 per cent). The size of villages also varies considerably. It is less than 200 persons in the hill states of north-eastern India, Western Rajasthan and Rann of Kuchchh and as high as 17 thousand persons in the states of Kerala and in parts of Maharashtra. A thorough examination of the pattern of distribution of rural population of India reveals that both at intra-State and inter-State levels, the relative degree of urbanisation and extent of rural-urban migration regulate the concentration of rural population.

You have noted that contrary to rural population, the proportion of urban population (31.16 per cent) in India is quite low but it is showing a much faster rate of growth over the decades. The growth rate of urban population has accelerated due to enhanced economic development and improvement in health and hygienic conditions.

The distribution of urban population too, as in the case of total population, has a wide variation throughout the country (Appendix–iii).

Activity

Compare the data of Appendix (iii) and identify the states/UTs with very high and very low proportion of urban population.

It is, however, noticed that in almost all the states and Union Territories, there has been a considerable increase of urban population. This indicates both development of urban areas in terms of socio-economic conditions and an increased rate of rural-urban migration. The rural-urban migration is conspicuous in the case of urban areas along the main road links and railroads in the North Indian Plains, the industrial areas around Kolkata, Mumbai, Bengaluru – Mysuru, Madurai – Coimbatore, Ahmedabad – Surat, Delhi – Kanpur and Ludhiana – Jalandhar. In the agriculturally stagnant parts of the middle and lower Ganga Plains, Telengana, non-irrigated Western Rajasthan, remote hilly, tribal areas of north-east, along the flood prone areas of Peninsular India and along eastern part of Madhya Pradesh, the degree of urbanisation has remained low.

Linguistic Composition

India is a land of linguistic diversity. According to Grierson (Linguistic Survey of India, 1903 – 1928), there were 179 languages and as many as 544 dialects in the country. In the context of modern India, there are about 22 scheduled languages and a number of non-scheduled languages.

Activity

See how many languages appear on a Rs 10 note.

Among the scheduled languages, the speakers of Hindi have the highest percentage. The smallest language groups are Sanskrit, Bodo and Manipuri speakers (2011). However, it is noticed that the linguistic regions in the country do not have a sharp and distinct boundary, rather they gradually merge and overlap in their respective frontier zones.

Linguistic Classification

The speakers of major Indian languages belong to four language families, which have their sub-families and branches or groups. This can be better understood from Table 1.2.

Religious Composition

Religion is one of the most dominant forces affecting the cultural and political life of the majority of Indians. Since religion virtually permeates into almost all the aspects of people's family and community lives, it is important to study the religious composition in detail.



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